

Swiss Re SONAR

New emerging risk insights



Every year, Swiss Re SONAR informs and inspires conversations about emerging risks, helping the insurance industry and its clients continue to build resilience.

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Emerging risk themes

Potential impact		0 – 3 years	> 3 years
High		  Beyond broken infrastructure – the cascading effects of natural catastrophes	
		  AI – unintended insurance impacts and lessons from “silent cyber”	
Medium		  Underfunding of public health – harmful to morbidity, mortality and GDP	  Climate change – an evolving threat to international security
		  Global supply chains – resilience against business interruption risk is weakening	
		  Big tech – a dependency risk	
		  Risky bets – democratising financial information through social media	
		  Cyber-enabled fraud – a new era for organised crime	
Low		  Smart drugs – wrong fix for cognitive enhancement?	  Recycling – when energy infrastructure becomes hazardous
		  Social isolation and loneliness – a growing health crisis	
		  No room left – competition for space generates infrastructure accumulation risks	
		  Alluring abyss – deep sea mining’s murky future	
Most affected business areas			
 Property lines  Specialty lines  Casualty  Life & Health  Financial markets incl. insurers’ assets  Operations incl. regulatory changes			

Trend spotlights



- FemTech: bridging the gender gap in healthcare
- Insurance fairness: challenge and opportunity
- Next generation nuclear

Foreword

Today, it can seem that every day delivers a new disaster or crisis, and more human suffering. The shadow of the pandemic still looms, health-wise and economically, and geopolitical tensions are at their peak, with new wars and frontiers drawn. Global society is becoming increasingly fragmented just when international collaboration is critical to face off the existential challenge of climate change.

We live in an age of poly-crisis, in an environment characterised by record-breaking temperatures, extreme weather events, war, terrorism and social discontent, not least due to increases in the cost-of-living. One crisis nourishes others, a chain reaction of greater uncertainty, risk accumulation and loss potential, both of human lives and of economic/financial value.

It is in times of adversity that the utility of the re/insurance industry as a stabilising force comes to the fore. Our proposition and capability is to provide a bridge when disaster hits, and to facilitate financial security and recovery. The question for re/insurers is not whether to worry about the state of the world, but how insurance solutions can help alleviate negative outcomes. In other words, how can we make the world more resilient?

Foresight helps inform preparedness, for us and our clients. Each year, Swiss Re SONAR presents insights into emerging risks, based on discussion with in-house experts, scientists, clients and industry peers. SONAR is about risks of the future, a vehicle to foster better understanding of those new or changing risks, their interactions and dependencies. SONAR does not claim the closing word on emerging risks. The aim is to stimulate engagement among all relevant stakeholders, including re/insurers, to help best prepare society for tomorrow's risks.

Welcome to SONAR 2024.



A handwritten signature in black ink, reading 'Patrick Raaflaub'.

Patrick Raaflaub
Group Chief Risk Officer

Introduction

The world undergoes constant change and this gives rise to “emerging risks”. These are new and changing risks that are difficult to quantify, yet can have significant impact on the insurance industry. Swiss Re identifies emerging risks by gathering signals and feedback from underwriters, client managers, risk experts and others across the Group, and also from external specialists and research institutions. With the annual SONAR report, published since 2013, we aim to foster dialogue with all stakeholders to better understand and manage emerging risks.

This year’s SONAR report features 13 emerging risk themes and three trend spotlights. The emerging risk themes are what could be new or changing risks, with both up- and downside potential for insurers. The “trend spotlight” items highlight contextual developments we deem relevant for the industry, without necessarily profiling a specific risk.

The emerging risk themes outlined in the SONAR report are based on early signals collected over the course of a year. They do not reflect industry-wide thinking with respect to emerging risks, nor do they necessarily cover the full list of associated topics currently on Swiss Re’s radar screen. The risks have been categorised according to their estimated time horizon, overall potential impact and by the lines of business that we think will be most exposed (see page 4).

For all the risk themes, we flag where there have been topic-relevant mentions in previous SONAR reports. Some emerging risks presented in this year’s report may never materialise, while others may form the basis for future risk pools. Similarly, some of the trends depicted may lose importance, while others may play an evolving and growing role in shaping the future business environment.

We begin the SONAR report with an overview of macro trends deemed by Swiss Re to be relevant for re/insurance markets in the coming decade. These trends and their grouping into demographic & social, political & economic, technological & natural, and competitive & business themes serve as a backdrop for the emerging risk themes and trend spotlights. The appendix explains all terms & definitions used in this report.

Per lines of business, the top emerging risk themes identified in this year’s SONAR are:

- for Property:
“Beyond broken infrastructure – the cascading effects of natural catastrophes”
- for Speciality:
“Beyond broken infrastructure – the cascading effects of natural catastrophes”
- for Casualty:
“AI – unintended insurance impacts and lessons from “silent cyber”
- for Life & Health:
“Underfunding of public health – harmful to morbidity, mortality and GDP”
- for Financial markets (including insurers’ assets):
“Risky bets – democratising financial information through social media”
- for Operations (incl. regulatory changes):
“Cyber-enabled fraud – a new era for organised crime”

Macro trends

Identifying and monitoring macro trends enhances our perception of what the future risk landscape could look like. Swiss Re curates a set of macro trends that we deem will be of high importance for the insurance industry over the coming decade. The macro trends portfolio informs Swiss Re's strategic priorities. Since 2015, we have publicly

shared our macro trends portfolio through SONAR. This year's portfolio of 27 macro trends is unchanged from 2023, when it had been reviewed and updated.¹ Below is an overview of the trends, followed by some of the current global focal points and emerging risk perspectives connecting this year's SONAR topics.

Demographic and social environment

- Shifting demographics and global aging
- Growing middle class in high growth markets
- Longevity & radical medical innovation
- Prevalence of mental health issues
- Mass migration & urbanisation
- Changing workplace and talent gaps
- Rising social inequality & unrest

Political and economic environment

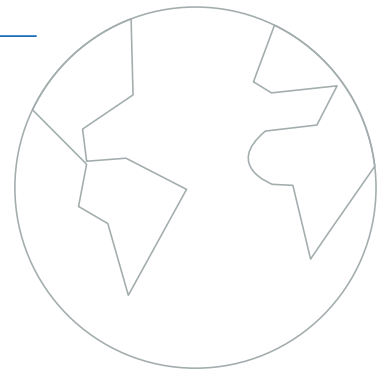
- Macroeconomic fragility
- Challenged globalisation
- Geopolitical & economic instability
- Rising interest rates and risk of persistent inflation
- Infrastructure funding needs

Technological and natural environment

- Addressing physical climate change risks
- Rising importance of biodiversity and ecosystem services
- Transition to a low carbon economy
- Expansion of digital & cyber risk
- Data as an asset
- Impact of generative AI
- Digital products and processes
- Disruptive digital technologies
- Autonomous transportation & robotics

Competitive and business environment

- Re/insurance value chain disaggregation and rise of alternative re/insurance providers
- Consolidation of platforms as a business model through strategic partnerships
- Regional champions going global
- Increasing digital customer interaction
- Increasingly litigious environment
- Rising importance of Environmental, Social and Governance (ESG)



Macro trends outlook – some current global focal points

Political and economic environment

The macro trends we highlight for the political and economic environment are currently in, what could be described as, “high-alert status”. Geopolitical instability manifests most prominently through ongoing wars, which have led to relative shifts in government spending priorities. Globally, military spending is surging,² attesting to the prevalence of perceived and real geopolitical instability. It also means less spending in also-important domains, such as (civil) infrastructure. Further, political tensions, open war and infrastructure funding gaps are increasing the risk of disruptions to supply chains (see “Global supply chains – resilience against business interruption risk is weakening” in this SONAR report, p.24).

Inflation persists, with large variations across the world. There is a trend to disinflation, but the rate at which prices head back to target depends on central bank monetary policy actions, these decisions in turn being driven by domestic economic growth dynamics.

National and parliamentary elections are being held in many countries in 2024. The US presidential vote in November takes centre stage but elections in India, the European Union and others will set domestic agendas for the years to come with, in turn, also international ramifications. Election outcomes will shape geopolitical coalitions and balance of power in ongoing conflicts, and will also influence willingness to collaborate on pressing issues such as climate change and the transition to a low-carbon economy.

Demographic and social environment

Persistent inflation continues to pressure households, business and government budgets around the world. Rising prices and funding gaps in healthcare systems (see “Underfunding of public health – harmful to morbidity, mortality and GDP” in this SONAR report, p. 12), are prolonging the stresses on care service provision in many countries that the recent pandemic gave rise to. There is currently an exodus of healthcare professionals to sectors, jobs and even countries that offer more attractive salaries and working conditions.

Demand for healthcare continues to rise as societies age, often outstripping supply. Affordability of healthcare services is a growing challenge, and there appears to be a widening discrepancy between what medicine and treatments could and can do for the few, and the practical healthcare possibilities available for the many. This adds to rising social inequalities among different age cohorts, genders and income groups. Acknowledging the inequalities can facilitate better assessment of currently

disadvantaged groups and improve the health outcomes for all (see “FemTech: bridging the gender gap in healthcare” in this SONAR report, p. 18). Isolation and loneliness are being recognised as growing problems in terms of societal well-being, resulting in both poorer physical and mental health, and impeding access to healthcare services (see “Social isolation and loneliness – a growing health crisis” in this SONAR report, p. 16).

Aging demographics remain a central macro trend for mature markets. Skill and talent gaps are being exacerbated by the large cohorts of the baby-boomer generation already or soon heading into retirement. The outcome is growing imbalance between the productive population and dependant demographics. Migration from countries with younger populations can be part of the solution. But immigration itself is a sensitive topic for many, causing social tension and giving some excuse for unrest.

Technological and natural environment

Climate risks present an existential threat to societies. 2023 was the hottest year on record and climate change translates into variations in the frequency and severity of some natural hazards, bringing higher economic losses for certain perils in some regions in the world.³ (see “Beyond broken infrastructure – cascading effects of natural catastrophes” in this Sonar report, p. 34). Over the longer term, climate change effects may also drive food and water shortages, and mass migration pressures that in turn could destabilise societies and international relations (see “Climate change – and evolving threat to international security” in this SONAR report, p. 22). Currently a focal impact area of concern is supply chain resilience (see “Global supply chains – resilience against business interruption risk is weakening” in this SONAR report, p.24).

The transition to a low-carbon economy remains high on the global political and economic agenda, and technologies to enable transition to renewable energy are being deployed more broadly. Wind and solar are developing at speed, and new forms of nuclear energy are also being pursued in many markets (see “Next generation nuclear” in this SONAR report, p. 42).

Energy transition comes with new risks. These include more demand and competition for the critical minerals and metals needed in batteries, and also for other crucial systems/tools that generate, store and distribute “green” energy. One area of potential take-off is deep-sea mining, but this too brings risks. Harvesting minerals from the seabed threatens maritime ecosystems and could cause environmental damage (see “Alluring abyss – deep sea mining’s murky future” in this SONAR report, p. 38).

Other trade-offs to consider relate to the future handling of end-of-life green technologies and recycling

procedures to prevent environmental damage (see “Recycling – when energy infrastructure becomes hazardous” in this SONAR report, p. 40). A growing density of green energy infrastructure and networks also entails competition for space to build and run both offshore and onshore infrastructure assets. Many of these have crucial network nodal points, heightening the risk of service interruption (see “No room left – competition for space generates infrastructure accumulation risks” in this SONAR report, p. 28).

The vulnerability of operating systems to technology dependency is central to macro-trend risks. That includes cyber risk, with ongoing digital innovation generating more and also new exposures (see “Cyber-enabled fraud – a new era for organised crime” in this SONAR report, p. 36). Loss accumulation potential and systemic risk are accentuated by concentration of service providers (see “Big Tech – a dependency risk” in this SONAR report, p. 26), and the power of digital platforms and communication networks to infiltrate all avenues of life. At the same time, social media is shaping decision making and can impact financial market volatility (see “Risky bets – democratising financial information through social media” in this SONAR report, p. 30).

Competitive and business environment

At accelerated speed, digitalisation continues to shape insurance business and related operations. The advent of artificial intelligence (AI) comes with many opportunities and a broad range of challenges, at various levels (see “AI – unintended insurance impacts and lessons from ‘silent cyber’” in this SONAR report, p. 46). In the field of generative AI, the release of new tools and updates offering new capabilities follow each other at speed. AI influences many categories of risk such as cyber and fraud (see “Cyber-enabled fraud – a new era for organised crime” in this SONAR report, p. 36), and also decision-making in compliance and regulation.

For the insurance industry, the range of opportunities and challenges from digitalisation involves insurers and the insureds. This two-way relationship entails complex and evolving stakeholder demands. Another expectation the insurance industry is faced with is greater adherence Environmental, Social and Governance regulatory standards, and discussion around transparency and fairness of insurance business vis-a-vis consumers (see “Insurance fairness: challenge and opportunity” in this SONAR report, p. 48).

¹ See Swiss Re SONAR: New Emerging Risk Insights 2023, p. 7ff.

² Stockholm International Peace Research Institute (SIPRI) reports a 7% surge for global military spending, reaching an estimate of 2.43 trillion USD in 2023, marking the most substantial annual increase since 2009.

³ *sigma* 1/2024: Natural catastrophes in 2023 | Swiss Re





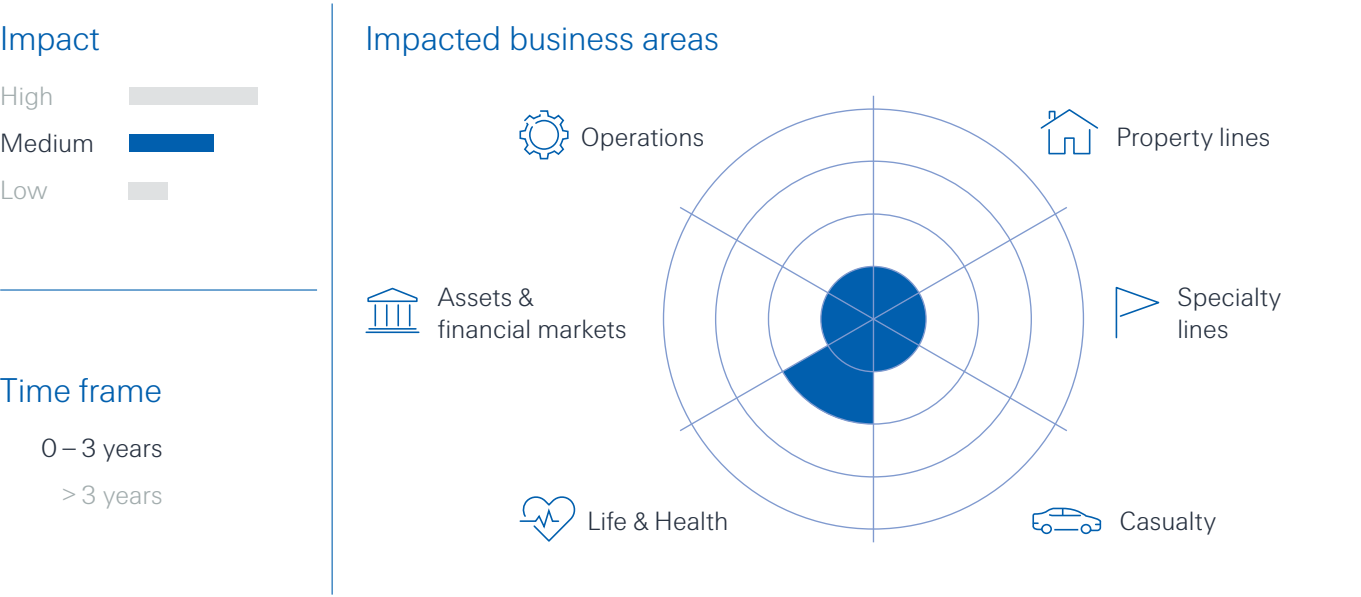
Demographic and social environment

Underfunding of public health – harmful to morbidity, mortality and GDP

A robust healthcare system requires countries to spend around 7% to 7.5% of gross domestic product (GDP). However, 132 countries spend less than his threshold.⁴ Over time, consistent underinvestment could lead to higher morbidity and mortality rates, potentially exacerbating future pandemics and also resulting in lower global GDP.

Potential impacts

- Delayed and/or inadequate care can lead to higher morbidity and mortality claims. In the case of a pandemic or severe endemic disease, a increase in mortality rates may be expected if healthcare systems crumble under overwhelming demand pressures.
- Greater number of medical malpractice claims can be triggered by insufficient care and/or human error.
- Health-related absenteeism and understaffing increases the risk of accidents, and incidents at industrial facilities can lead to property losses. If offsite impacts occur, casualty losses and workers' compensation/employers liability claims may be triggered.
- Economic disruption on same or even worse scale than that caused by COVID-19 will diminish GDP growth and with it insurance market growth.
- If mortality in a pandemic is high, there is potential for population shrinkage, affecting demand for insurance.



Exodus in healthcare

Under-financing of healthcare systems and the impacts thereof are evident across all economic spectra, be it low-, middle- or high-income countries.⁵ In the UK, for instance, insufficient funding has brought the National Health Service (NHS) to breaking point.⁶ A critical consequence of this is chronic shortage of doctors and nurses. Over the past 4 years, approximately 20 000 doctors and 38 000 nurses have left the NHS, with many exploring overseas options in countries where employment in the health sector is better compensated.⁷ With the exodus, attracting new staff in a home market is a challenge. The ripple effects of healthcare workers moving from

low-income to high-income countries are many. These include a weakening of the resilience of healthcare systems in poorer countries that continue to struggle to provide a robust care infrastructure for the local population.⁸ And among other challenges, staff shortages can lead to delays in appointments and treatments, as already experienced in the UK and other developed countries with sizeable L&H markets.⁹

Shortcomings of infrastructure or supplies can hurt

In addition to staffing issues, a resilient healthcare system requires adequate infrastructure, including hospitals and medical equipment. As global temperatures continue to

rise, a lack of climate-resilient healthcare infrastructure may become a more pressing issue. Essential services such as water, sanitation and electricity supplies may be compromised under more extreme climate scenarios that generate heightened risk of frequent flooding and other disruptive events (eg, an outbreak of dengue fever).¹⁰ In cases of extreme heat and related droughts, water cuts may be necessary to regulate the supply of hydroelectric power. In addition, water temperatures might be too high to cool nuclear reactors resulting in lower energy output to hospitals and other industry sectors, and households.¹¹

The availability of robust power supplies are also important if healthcare systems are to capitalise on the benefits that new digital technologies offer. This message is reinforced by the World Health Organization (WHO) in its *Global Strategy on Digital Health 2020 – 2025*,¹² which advocates the use of latest digital-based technologies to enhance healthcare provision. Without reliable power, the software and hardware that are fundamental to digital services cannot operate effectively.

Funding gaps affect societies and economies

Ultimately, despite widespread acknowledgement and understanding of the roles of a healthcare system, and also consideration of the needs of ageing populations and related health vulnerabilities, in many countries levels of funding consistently fall well short of need.

Within the framework of the Sustainable Development Goal 3.8 on Universal Health Coverage, the WHO remarks that “the world is off track to make significant progress towards universal health coverage (UHC) by 2030 as improvements to health services coverage have stagnated since 2015, and the proportion of the population that faces catastrophic levels of out-of-pocket health spending has increased.”¹³

The lack of funding for healthcare – resulting in limited treatment and stretched resources – has arguably already shown in the increase in mortality rates during the

pandemic and delays in clearing healthcare backlogs.¹⁴ For L&H insurers, a potential worsening of morbidity and mortality trends is critical for risk assessment and underwriting outcomes. In the event of an epidemic or pandemic, even more so, as the insured population cannot be isolated from the prevailing health crisis impacting society at large.

Related articles in SONAR

- “How to better prepare for the next pandemic”, SONAR 2022
- “Beyond pandemic – longer-term emerging risks for insurance”, SONAR 2021
- “Income inequalities – pandemic hurts the middle class”, SONAR 2021
- “The fragility of public healthcare”, SONAR 2020
- “Global pharmaceutical supplies – breaks in the chain?”, SONAR 2020
- “Paradise lost – the price of ecosystem services”, SONAR 2018
- “Rising pandemic risk”, SONAR 2015
- “The antibiotic boomerang”, SONAR 2015

- ⁴ M. McConnell, [Governments should step up on public health care spending](#), Human Rights Watch, 20 Sept. 2023.
- ⁵ A country’s healthcare expenditure will be influenced by its demographic distribution and the underlying health and vulnerabilities at a population level. Factors such as healthcare inequity and the balance between preventative medicine and treatment also play significant roles in determining spending priorities.
- ⁶ J. Reed, [NHS in England facing worst staffing crisis in history, MPs warn](#), BBC 25 July 2022.
- ⁷ Z. Kahn, [The emerging challenges and strengths of the national health Services: a physician perspective](#), Cureus 15, May 2023.
- ⁸ P. Adepoju, [Healthcare workforce shortages exacerbated by poaching from the global South](#), Nature Medicine 30, Jan. 2024.
- ⁹ [NHS backlog data analysis \(last update 12 April 2024\)](#); WHO, The health workforce crisis in Europe is no longer a looming threat – it is here and now, Bukarest, 22 March 2023.
- ¹⁰ [Climate Change and Health \(who.int\); Visual summary – Lancet Countdown](#)
- ¹¹ [Europe’s Scorching Summer Puts Unexpected Strain on Energy Supply, The New York Times, 18 Aug 2022](#).
- ¹² [WHO, Global Strategy on Digital Health 2020 – 2025](#).
- ¹³ [World Health Organization and International Bank for Reconstruction and Development/ The World Bank, Tracking universal health coverage: 2023 global monitoring report, Geneva 2023](#).
- ¹⁴ [Swiss Re Institute, The future of life expectancy, May 2023](#).



Smart drugs – wrong fix for cognitive enhancement?

Micro-dosing of psychedelic drugs like LSD¹⁵ or magic mushrooms, or “enhancing” prescription drugs with boosters for better concentration, memory and productivity at work, has become a trend for adults in the workplace.¹⁶ But what if usage does not achieve better results?

Potential impacts

- Improper management decisions made under the influence of cognitive enhancement drugs can lead to losses in property, casualty, professional indemnity and even D&O insurance.
- The effects of long-term exposure to psychedelic drugs are still poorly understood and could lead to rising claims in product liability or professional indemnity insurance.
- Usage in high risk occupations can lead to insurance claims. Examples are operators of hazardous facilities, workers in healthcare or financial market traders taking erroneous decisions under the influence of drugs. Such mistakes can also generate financial market volatility.
- Accidental or intentional misuse of drugs can lead to sickness and in extreme cases, death among working professionals.

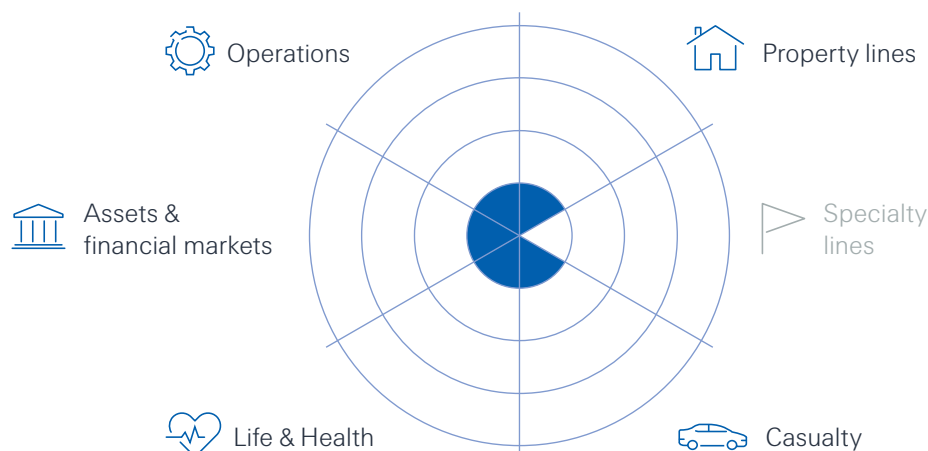
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



In 2023, Science Magazine published a study¹⁷ relating to three “smart drugs”:¹⁸ Methylphenidate and dextroamphetamine used for the treatment of attention deficit hyperactivity disorder (ADHD), and Modafinil used for narcolepsy. The drugs were administered to participants together with a placebo for control. The participants had to solve a problem that required cognitive resources like working memory and concentration. The results were counter-intuitive to the drugs’ intended outcomes. There was a decline in the quality of effort on the part of the participants to solve the problem. And when they took the placebo, the participants performed better than when they had taken one of the drugs.

These outcomes raise important questions, particularly in the context of the people likely to use smart drugs. This includes persons who work long hours in high-performance stressful environments, such as surgeons. An overview study published in the UK¹⁹ found that 8.9% of surgeons had used prescription or illegal substances at least once for cognitive enhancement. The study closes by saying the numbers are likely to rise as smart drug use becomes increasingly popular, including (already) among students. Stressful work environments are not unique to surgeons. Bus drivers, pilots, workers in high-risk environments like chemical plants or the health care sector are just a few examples. Many of these people may consciously or unconsciously combine prescribed and/or smart drugs

with food supplements,²⁰ which are also becoming more popular. These may potentially bring unintended side effects as some may be contaminated with other drugs or ingredients.²¹

Unapproved substances

The effect may even be more pronounced if users obtain illegal drugs not yet approved for micro-dosing, such as LSD, ketamine or magic mushrooms,²² the use of which is reportedly increasing.²³ These substances have become more popular in management circles as means to enable new insights and increase productivity.²⁴ What is often forgotten is that news insight are not necessarily always correct ones, and hence also not helpful for decision making.

Adequate decisions are key in high-risk, high-impact occupations. If many professionals turn to smart drugs to enhance performance and the effect is actually opposite, there could be many negative outcomes. In high-risk environments these outcomes are often events that destroy property, lead to bodily injury or loss of life, and financial losses. In high-impact occupations like management, bad decisions made under the influence of smart drugs could harm company performance, which in turn can lead to losses in professional indemnity and/or D&O lines of business.

Lastly, there may still be currently unknown long-term effects of many substances, especially those where use in the past was mainly illegal and recreational. A study published in 2023 indicated that males who take ADHD

medication can encounter issues with testosterone levels later in life.²⁵ Insurers should be on the lookout if claims involving smart drug use are increasing, particularly for jurisdictions, where the legal framework makes access to smart pills relatively easy.²⁶

Related articles in SONAR

- “Eyes wide shut – the world is sleeping less”, SONAR 2018
- “Faking it – the business of counterfeit medicine”, SONAR 2018
- “Nutraceuticals”, SONAR 2016
- “Lifestyle drugs”, SONAR 2015

¹⁵ [LSD \(from German Lysergsäurediethylamid\) stands for lysergic acid diethylamide a synthetic psychedelic drug.](#)

¹⁶ L. Cameron et al., [Psychedelic microdosing: Prevalence and subjective effects](#), J Psychoactive Drugs, 52 (2020 Apr – Jun).

¹⁷ E. Bowman et al., [Not so smart? “Smart” drugs increase the level but decrease the quality of cognitive effort](#), Science Advances, June 2023.

¹⁸ [Smart drug | Definition, Types, & Facts | Britannica.](#)

¹⁹ L. Thevapalan et al., [Smart drugs among surgeons: future cause for concern?](#), British Journal of Surgery 110, 2023.

²⁰ A. Cowan et al., [Trends in overall and micronutrient-containing dietary supplement use in US adults and children](#), NHANES 2007 – 2018, Journal of Nutrition 2023.

²¹ M. Amidzic et al., [Notifications and health consequences of unauthorized pharmaceuticals in food supplements](#), Pharmacy 2023.

²² D. De Gregorio et al., [Hallucinogens in mental health: Preclinical studies on LSD, Psilocybin, MDMA, and Ketamine](#), Journal of Neuroscience 2021.

²³ [Global Drug Survey, London.](#)

²⁴ K. Grind and K. Bindley, [Magic Mushrooms, LSD, Ketamine: The drugs that power Silicon Valley](#), Wall Street Journal 27 June 2023.

²⁵ G. Ostiek-Wille et al., [Prolonged use of ADHD medications is associated with testosterone hypofunction – results from a national claims database analysis](#), Int. Journal of Impotence Research 2023.

²⁶ K. Campanini, [“Tiktok told me I have ADHD”: Regulatory outlook for the telehealth revolution](#), Washington Journal of Law, Technology & Arts 18, 2023.; Marks, M. et al., [Microdosing Psychedelics Under Local, State, and Federal Law](#), 103 Boston University Law Review 573 (August 18, 2023).



Social isolation and loneliness – a growing health crisis

Social isolation and feelings of loneliness are being linked to increased health risks such as depression, heart conditions and dementia, among others. Globally and across age groups, the incidence of isolation and loneliness is on the rise.

Potential impacts

- Social isolation and feelings of loneliness can aggravate the mental health burden in disability protection, and increase risk of suicide (impacting life insurance).
- Earlier and greater need for treatment of mental health conditions impacts on long-term care insurance. Co-morbidity with a range of other chronic health issues affects all L&H insurance.
- Feelings of loneliness can have detrimental effects on employee productivity and lead to increased absenteeism.²⁷ The feelings may also contribute to errors and safety hazards in the workplace.

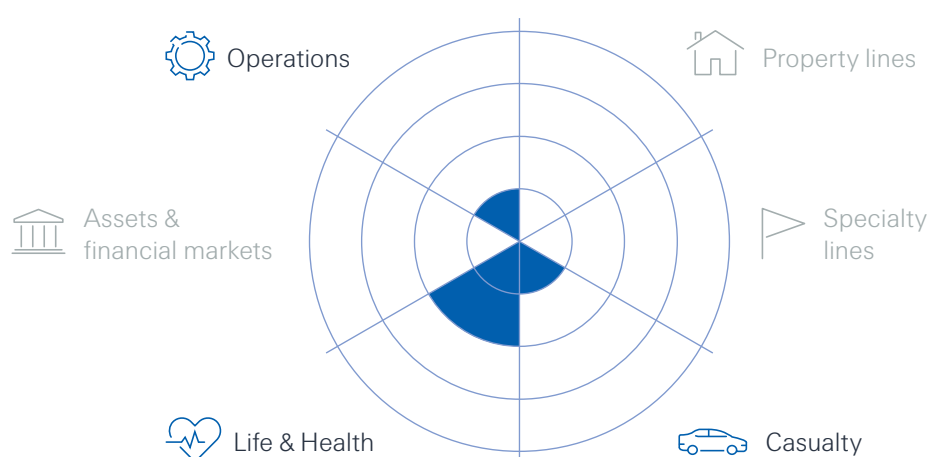
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



Social isolation and loneliness affect many...

Research indicates that social isolation and loneliness are a global problem. According to the World Health Organization (WHO), insufficient social contact and feelings of loneliness are experienced by one in four adults. Additionally, it is estimated that at least 5 – 15% of adolescents feel lonely. In the US, one study suggests that a third of the over-45 population has experienced feelings of loneliness. Certain demographic groups appear to be particularly exposed, including those on low incomes, young adults, the elderly, those living alone or single parents, people suffering from chronic disease or disability, and immigrants facing language barriers and/or discrimination.²⁸

... and are harmful

Social isolation can be broadly characterised in terms of contact frequency, quantity of relationships, and/or density

of interpersonal networks. Marginal social contact may also result in insufficient access to support in times of physical or mental crises, and other situations when personal contact, access to emergency services and/or welfare institutions are needed. Social isolation has been linked to an increased likelihood of health complications, accidents, malnutrition and other factors that can compromise overall well-being.²⁹

Loneliness is a subjective feeling of isolation that affects personal well-being. Perceived and objective social isolation can be damaging and are often intertwined.³⁰ A growing body of research links social isolation and loneliness to an increased risk of depression, anxiety, addiction, self-harm and suicidal tendencies, all these in addition to increased incidents of type-2 diabetes, dementia, heart disease and strokes.³¹ Estimates suggest that social isolation and loneliness lead to, for instance, a

50% increase in the risk of onset of dementia, and around a 30% increase in the likelihood of heart conditions.³² Collectively, these factors can contribute to higher morbidity rates and premature death. Studies have also suggested that the mortality risk from loneliness and social isolation may be comparable to that of cigarette smoking, alcohol, physical inactivity and obesity.³³

The pandemic made things worse

The COVID-19 lockdown and social distancing measures impacted the personal relationships and mental health of many individuals, irrespective of age. Researchers continue to investigate both the short and long-term mental health impacts of the pandemic.³⁴ In May 2023, the US Department of Health and Human Services raised the alarm on an “epidemic of loneliness.”³⁵ Acknowledging the adverse impacts that the pandemic had on social connections, in November 2023 the WHO established the new *Commission on Social Connection* to “address loneliness as a pressing concern to health.”³⁶

With increasing urbanisation and a shift in socioeconomic trends such as the rise of single-person households and anonymous living in apartment blocks/neighbourhoods, social isolation and loneliness can be expected to persist. In ageing societies, and with the large cohort of baby boomers retired or retiring, the proportion of the working-

age population capable of providing the required professional or private care services is diminishing. As demand exceeds supply, social isolation and feelings of loneliness could be further exacerbated.

Related articles in SONAR

- “COVID-19 – the longer-term health burden”, SONAR 2021
- “Out of sight, out of mind – mental health issues among the young”, SONAR 2020

²⁷ [The Loneliness Epidemic Persists: A Post-Pandemic Look at the State of Loneliness among U.S. Adults | Cigna Newsroom \(thecignagroup.com\) \(visited April 2024\).](#)

²⁸ [CDC, Health Risks of Social Isolation and Loneliness, 30 March 2023.](#)

²⁹ [The hurt of loneliness and social isolation. Nat. Mental Health 2, 255–256 \(2024\).](#)

³⁰ Taylor HO, Cudjoe TK, Bu F. et al., [The state of loneliness and social isolation research: current knowledge and future directions](#), BMC Public Health, 2023;23:1049.

³¹ S. Sidik, [Why loneliness is bad for your health](#), Nature News Feature, 3 April 2024.

³² N. Valtorta et al., [Loneliness and social isolation as risk factors for coronary heart disease and stroke: systematic review and meta-analysis of longitudinal observational studies](#), Heart 2016.

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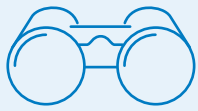
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FemTech: bridging the gender gap in healthcare

Historically women have been under-represented in health research. Gender medicine is rectifying this through sex-sensitive research and treatment. The gender gap has become big business with start-ups and private equity firms racing to provide personalised female care. But at what cost?

Gender inequality in healthcare

Gender discrimination in medical research has long been a reality. For example, typically male mice are used in early phase clinical trials as researchers think/have thought that the female hormone cycle can impact test results. The same biases reflect in the under-representation and underfunding of woman-centric research.³⁷ The outcome of an one-size-fits-all approach based on male-centric trials is incomplete knowledge of how gender, as a biological factor, impacts treatments and medicines that come on to the market.³⁸ Experience has shown that women experience adverse reactions to medicines twice as often as men, and in most cases suffer worse side effects too. This could be explained by women often being under-represented at all levels of clinical trials.³⁹

FemTech: the new buzz word

In recent years, research focus has become more gender specific.⁴⁰ This includes universities offering advanced trainings in gender-specific medicine.⁴¹ As inequalities are addressed, we see an increase of females in medical research, and new products and technologies being developed to positively change women's healthcare. In the US, investments in companies focusing on female healthcare – FemTech – rose from USD 0.8 billion in 2021 to USD 3.3 billion in 2022.⁴² Projections put the global FemTech market at USD 75 billion by 2025.⁴³

Among others, FemTechs provide digital personalised care, telemedicine, one-on-one support services and access to a peer community. So far investments have mostly been in obstetrics, gynecology and fertility services, but the scope is set to broaden. There are investments in new technologies like soft-tissue reconstruction techniques, detection tools for breast cancer and instruments to improve surgery experience and outcomes in female-centric procedures like caesarian sections and treatment for pelvic floor disorders.⁴⁴ Consumers of FemTech services are not only those who identify as female. Transgender and non-binary individuals also seek reproductive health, menstruation, fertility tracking, hormone therapy-related and other services.

Giving away your personal data

Gynecology and fertility devices are typically digital apps and trackers. Users track their health and well-being on such apps, which involves sharing information about their menstruation cycles, sleep patterns and stages of pregnancy etc, private information that is rarely shared outside a circle of close family and friends.

And therein lies the rub. A recent study found that it is not only the operators of certain popular apps that have access to sensitive data. So too do third-party tracker agents, who can collect user data from the apps and share them with other third parties like advertisers, analytics providers and social media platforms.⁴⁵ One aspect is how secure the data is and the potential individual impact of a data breach if sensitive information is leaked. On an individual account, a data leak could be very harmful given, for instance, recent political developments in the US to limit abortion rights and global backlash against gender and women's rights.⁴⁶ Another personal risk aspect is when a state has legal access to electronic data related to individuals, for instance to monitor women considering abortion.⁴⁷

Collecting personal data to target individuals with personalised online advertising is lucrative business. By consenting, users relinquish control – sometimes unwittingly – over how their data is used.⁴⁸ The consequences are manifold and potentially invasive. For example, a recent mother who uses an app may start to receive advertisements for weight-loss services. Such marketing can pressure the mother to live up to standards deemed socially desirable, to the detriment of her mental health and overall well-being.

Addressing gender health inequality gaps should be a priority in medical R&D, but should not come at the cost of exposing women to further personal risks.

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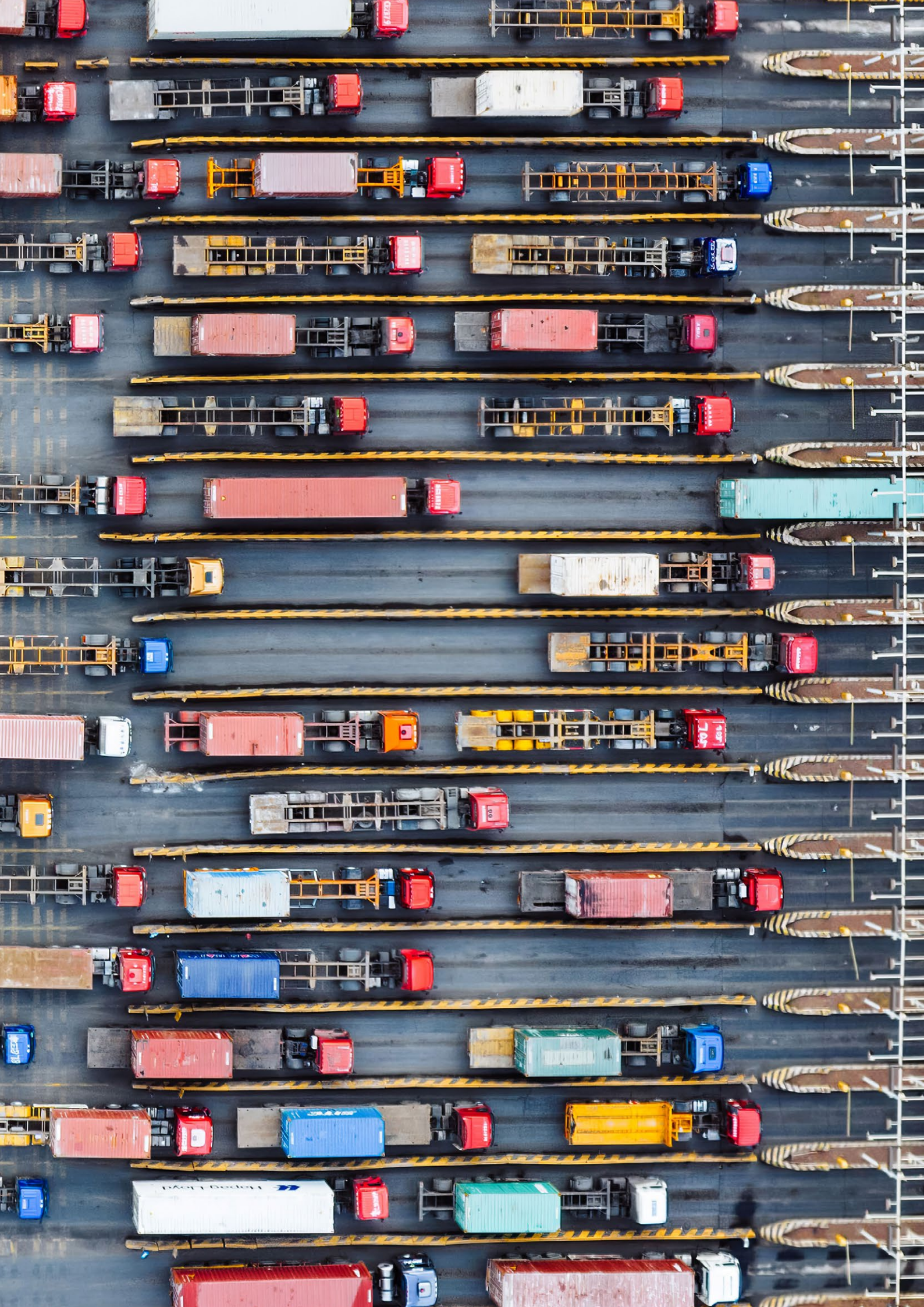
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Political and economic environment

Climate change – an evolving threat to international security

Climate change impacts are not a direct root cause of civil unrest, terrorism or armed conflict, but they are recognised as destabilising factors.

Potential impacts

- Domestic and international conflict can dampen economic growth and overall demand for insurance.
- For business generally, operational challenges posed by climate change include potential supply chain disruptions.
- Geopolitical uncertainties and the evolution of climate change dynamics will prompt shifts in the composition of demand for insurance. Demand for climate peril (eg, drought, flood and tropical cyclones) covers will likely rise, and so too will demand for political risk insurance.
- There will likely be more demand for strike, riots and civil commotion insurance also, although an actual rise in civil unrest could challenge the insurability of such risks.
- In the case of military conflict, insurers' liability and asset portfolios could come under stress, ultimately increasing the cost of capital.
- Strategic interest and economic activities in the Arctic region – ranging from trade (marine), to fishing, mining and tourism – is likely to grow as warmer temperatures continue to melt icebergs, opening up new travel routes and risk pools.

Impact

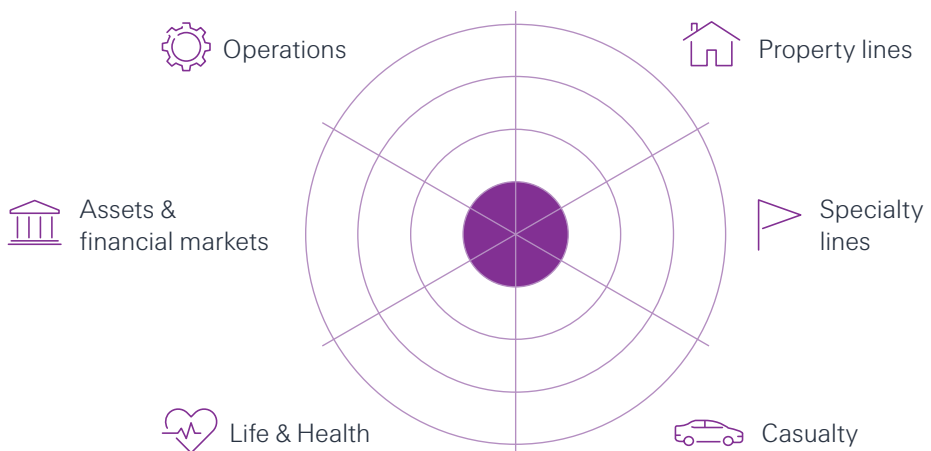


Time frame

0 – 3 years

> 3 years

Impacted business areas



Climate change impacts take many forms,...

Beyond the physical risks posed by climate-change induced variations in the patterns of severe weather and other natural peril events, the impacts of warming temperatures are manifold. They include, for example, the contribution of changing weather patterns in driving global food insecurity to record levels. In 2022, a quarter of a billion people in 58 countries faced acute food insecurity, and 35 million people faced starvation due to conflicts and an intensification of severe weather events.⁴⁹ At the same time, the world faces a fresh water crisis, experts say, with demand for fresh water forecast to exceed supply by 40% by 2030.⁵⁰ Drivers of the

anticipated water shortages include misuse and pollution of water resources, and changes to the global hydrologic cycle on account of climate change.

Climate change can also lead to loss of agricultural output, impacting rural household livelihoods and prompting mass migration. The UN estimates that by 2050, climate change could force 216 million people in six regions to move within their own countries.⁵¹ Seventy percent of the world's most climate-vulnerable countries are also among the most socially and economically fragile.⁵² With severe weather events occurring more frequently, inconsistent precipitation patterns and rising

sea levels, countries face increasing risk of civil disorder due to potential food scarcity and higher prices, competition for natural resources and mass migration.

... including undermining peace and global security

Climate change also has implications for peace and security,⁵³ not least as a destabilising factor that has contributed to the growth of extremist organisations.⁵⁴ Water stress serves as an example where extremist organisations leverage water to control supply and provide services to gain followers.⁵⁵ There are also examples when such organisations have offered work opportunities to those with alternatives to earn a living.⁵⁶

Further, several studies indicate an increase in general crime as temperatures rise. For example, one study reports a 22% increase in social media online hate messages across the US when temperatures rise above a 12 – 21°C window.⁵⁷ This correlation between higher temperatures and hate crime and others could further trigger social tensions within cities and countries.

In a different part of the world, melting icebergs in the Arctic Circle is opening up new trade routes and access to new resources. At the same time, however, this is triggering tension between states over influence and control in the region, leading to a growing risk of military escalation.⁵⁸

Climate change risk assessment

The cascading non-linear effects of climate change are challenging to project over the long term in general, and with respect to security developments.⁵⁹ Nevertheless,

from a security angle, climate change risk assessment is crucial at global and country levels. It is also vital for enterprises as part of business continuity planning. A company needs to review its operations for all potential disruption scenarios that changes in climate may present, beyond the physical risks posed by natural perils themselves.

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Global supply chains – resilience against business interruption risks is weakening

Post COVID-19, the security of supply chains was high on management agendas. Two to three years later, cost reduction seems to have taken centre stage again.⁶⁰ Less investment in supply chains makes them less resilient to shocks, be those natural catastrophes, unforeseen technology outages or political events, and increases the risk of business interruption-related losses.

Potential impacts

- Business, contingent and non-damage business interruption loss ratios could rise as less resilient supply chains lead to more business interruptions.
- The risk of D&O claims increases if there is less investment in supply chain resilience, resulting in disruption to business operations. This could be considered as gross negligence or even intentional neglect by management.
- Morbidity and mortality rates can increase if deliveries of medical supplies are interrupted.
- Longer disruptions (eg, next pandemic, civil unrest in supplier countries, blockage of major trade routes, volcanic eruptions impacting air freight, earthquakes/tsunamis impacting major economic regions) can lead to economic and insurance market downturn.

Impact

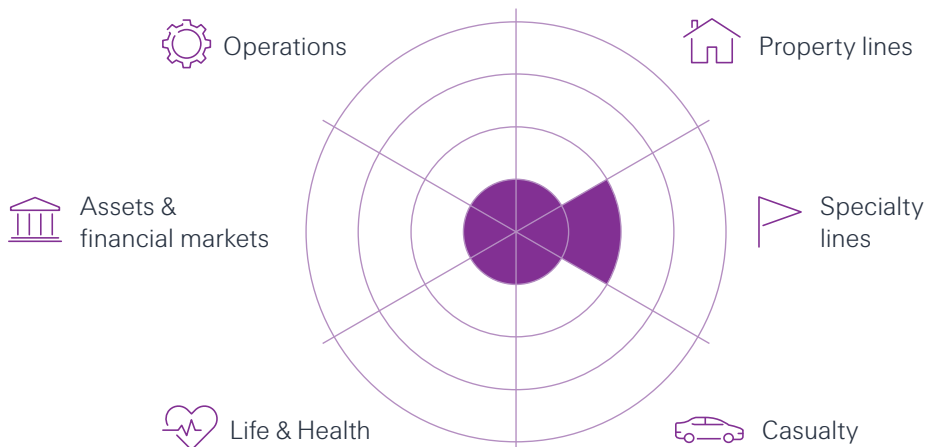


Time frame

0 – 3 years

> 3 years

Impacted business areas



As the COVID-19 pandemic experience showed, supply chains can be heavily disrupted, bringing long-lasting implications.⁶¹ Supply chain resilience is preventing breakdowns in delivery value chains, and the ability to quickly recover, rearrange and/or mobilise alternative delivery routes. Safeguarding supply chains can help companies maintain market share and profits, which a narrow focus on cutting spending on supply chains cannot. Looking back at former SONAR reports, supply chain risk has been a recurring theme. The majority of the articles have focused on external risk drivers like natural catastrophes or politics (ie, disruptive

events that are beyond direct influence of supply chain management). For many of these drivers the current outlook is negative.

Political risks

Besides the headline conflicts like Ukraine or Gaza, the Red Sea or the future of Taiwan, the political environment in many parts of the world has become less stable. The *Fragile State Index*⁶² for 2023 indicates fragility has spread from poorer countries to the more affluent parts of the world. The inference is that key supply routes around the globe are likely to become less secure.

Nature-related risks

Data indicates that the amount of insurance property owners buy against natural perils is increasing each year.⁶³ More frequent extreme weather events,⁶⁴ earthquakes and volcanic eruptions⁶⁵ can impact production facilities and transport routes.

Technology risks

While digitalisation can provide efficiency gains in supply chains, it also presents areas of systemic vulnerability. Emerging vulnerabilities require constant updates to hardware, software and procedures, and trained staff to keep systems safe and running.⁶⁶

Infrastructure risks

Supply chain operations depend on infrastructure. Energy, water and power, and also health infrastructure, which becomes especially important in the case of a man-made or natural disasters, or a pandemic. Still, according to the G20,⁶⁷ there is a big gap between investment need and actual spend. The result could be less resilient supply chains.

Economic risks

Economic drivers can impact investments in supply chain infrastructure. In return, supply-chain constraints can impact the economic outlook. This is why the Federal Bank of New York created its *Global Supply Chain Pressure Index*.⁶⁸ If any of the drivers cited before coincides with an already stressed supply chain, the fallout for businesses can be significant.

Given the negative outlook on many external risk drivers and the pandemic experience, one would expect that firms have supply chain resilience on the top of their agendas. According to a recent survey by PWC,⁶⁹ today this is not the case. Only one-third of executives cited increasing supply chain resilience with more investment as a top objective. Instead, the top priorities of executives are operational efficiency and cost reductions.

Insurers should acknowledge this short-term view when underwriting business interruption and contingent business interruption risks. This especially as risk drivers can interact with each other. If drivers accumulate, the losses incurred by insurers can increase considerably. An example would be a pandemic coinciding with a period of drought. A lack of water and then also electricity could cause disruptions to supply chains and business operations. There could also be a decline in productivity due to absence of workers on account of ill health.

Related articles in SONAR

- “Bird flu – wake-up call for outbreaks of animal diseases?”, SONAR 2023
- “Sustainable recycling – emerging liabilities in the circular economy”, SONAR 2023
- “Climate change, trade and geopolitics – new risks in the Arctic”, SONAR 2023
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Big Tech – a dependency risk

Big Tech is omnipresent. Dominance fosters dependence, but who's accountable?

Potential impacts

- Shutdown of services from one or more of the Big Tech providers can disrupt the operations of corporates and other organisations in all sectors. In turn, insurance covers may be triggered. Insurers' own operations may also be impacted, and financial services could be affected on the investment side.
- Issues around algorithms offered as a service could lead to professional indemnity, product liability or employment practices liability claims due to errors or bias in data. The draft European Product Liability directive defines software as a product, thereby increasing product liability exposures.
- The increased feeding of data into open-source AI or social media platforms can potentially result in breach of privacy and security, intellectual property and duty claims. This could increase cyber, D&O and professional indemnity exposures.
- Heightened scrutiny of and more stringent regulations for Big Tech with respect to ethical standards, alongside societal and political influence, could lead to an increase in D&O claims.
- Overuse of digital platforms can be distracting and lead to omissions in critical safety functions (eg, system steering roles) and accidents. This could impact liability lines of insurance business: motor (car accidents) and general liability, professional indemnity, and employer liability/workers' compensation (workplace accidents). Increasingly, the companies behind digital platforms may be held liable for the negative effects – mental and physical – experienced by users.

Impact

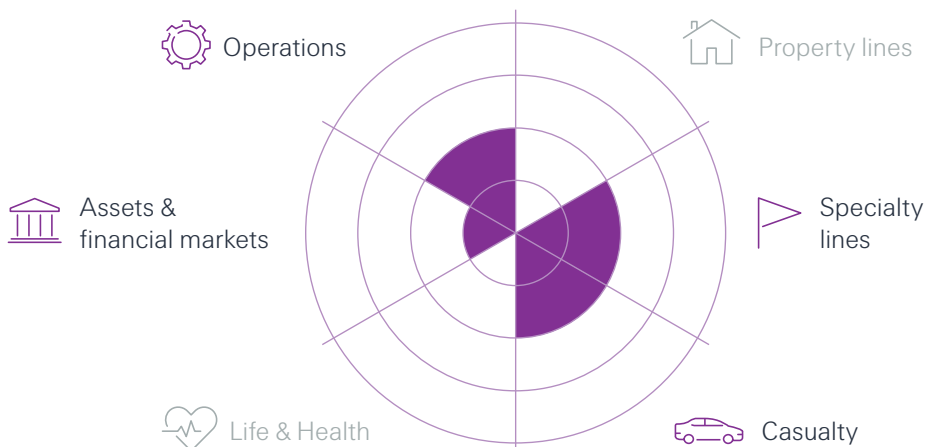


Time frame

0 – 3 years

> 3 years

Impacted business areas



Global dependency

Big Tech firms such as the so-called Big 5 – Alphabet (Google), Amazon, Meta (Facebook), Microsoft, and Apple – have become omnipresent in daily life. Their global reach spans millions of users worldwide, they have access to Big Data and they are quasi-monopolistic.

Google is utilised for 91.5% of searches⁷⁰ and Google Chrome has a 64.4% share of the browsing market.⁷¹ Apple's iPhone has a market share of 27.1%,⁷² while Facebook and Instagram (owned by Meta) claim a 76%

share of social media market.⁷³ Amazon dominates US e-commerce with a market share of 37.6%,⁷⁴ and globally runs second only to Alibaba.⁷⁵ Microsoft computers run 73% of desktop operating systems worldwide.⁷⁶

Amazon, Microsoft and Google host critical business operations on their cloud services for many companies. Amazon Web Services is the leader with a 31% market share in the fourth quarter of 2023, followed by Azure (Microsoft, 24%) and Google Cloud (Alphabet, 11%).⁷⁷ Technology research and consulting company Gartner

has estimated that public cloud end-user spending in 2023 will reach nearly USD 600 billion, up 22% from the previous year.⁷⁸

Interconnected services

With the steady increase in use of cloud services provided by just a few Big Tech firms, the world economy and society face a dependency risk.

The services provided by Big Tech include search engines, browsers, smart phones, social media, e-commerce, software, hardware, operating systems, cloud services and most recently generative artificial intelligence, and these services are becoming increasingly interconnected. The main cloud service firms are building several layers of redundancy by separating infrastructure centres in various ways (eg, on the software side, physically and geographically). Nevertheless, occurrence of a disaster event due to human error,⁷⁹ natural catastrophe or cyberattack, resulting in a long-running shutdown of services provided by different companies and the public sector is an ever-present risk.

In addition, in recent years, the technology industry overall and the Big 5 have been hitting the headlines due to concerns around the addictiveness of social media and other software services such as gaming, the political influence and polarisation that social media platforms can facilitate, and data privacy. For instance, social media has come under more scrutiny for allegedly

undermining democracies and human rights, and for facilitating societal polarisation, this in turn leading to ethnic violence.

Given their global reach and influence, what will happen, and who will be accountable should one or more of the Big Tech companies fail?

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- “Hooked online – anxiety and distraction from social media binging”, SONAR 2023
- “AI getting hacked – systemic vulnerabilities of a booming technology”, SONAR 2023
- “Ethics in digital nudging”, SONAR 2021
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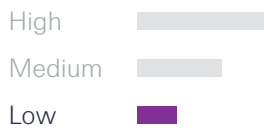
No room left – competition for space generates infrastructure accumulation risks

The ground space required for off- and near-shore infrastructure installations is expected to increase fivefold by 2050.⁸⁰ Onshore installations are expected to grow also.⁸¹ Competition for space to build and run infrastructure assets can lead to an accumulation of business interruption and national security risks.

Potential impacts

- Damage to infrastructure gives rise to property, business interruption and contingent business interruption claims.
- If a service provider overlooks an exposure, there could be professional indemnity losses.
- D&O claims are possible in cases of severe negligence.
- Longer interruptions can dampen economic growth and the outlook for insurance markets.
- In case of damaged oil pipelines, environmental liability claims will likely be triggered.

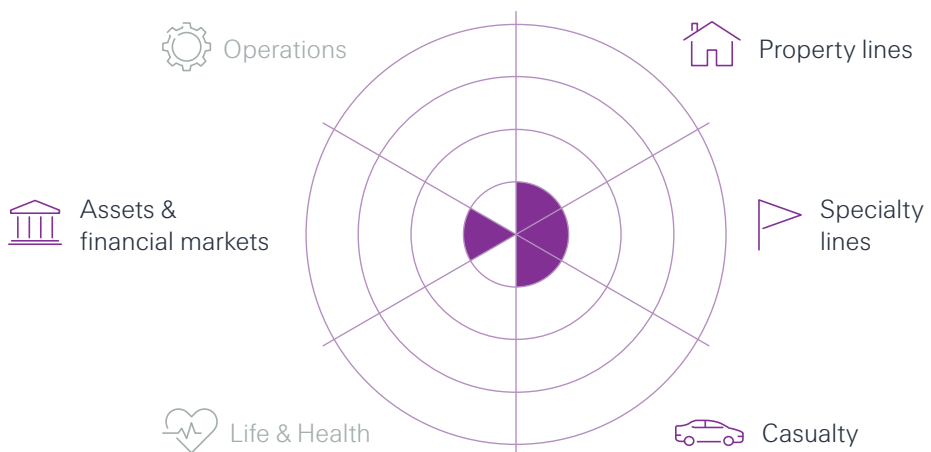
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



The race for space is on, off- and onshore. The expectation is that by 2050, the physical surface ground area required to build and operate marine aquaculture, and offshore wind and oil & gas infrastructure, will increase fivefold. Not included in this estimate is the space needed for structures that protect the natural environment and support the tourism industry. Also not included are floating solar farms, desalination and wave & tidal energy plants, submarine cables, pipelines, seabed mining equipment and installations used for military purposes.⁸²

A fine line

Near-shore, where today most damage events happen, will become especially crowded.⁸³ Having more infrastructure in the same marine space heightens the risk of damage to

different installations. This includes potential for third-party damage and cascading effects, when one specific installation connects to one or more other networks.⁸⁴

For example, power and data networks. If part of a system fails, the energy traffic is normally routed to other parts of the network. If too much power flow goes to one network, the latter can become overloaded and fail, which in turn could render an entire network obsolete until repair work is completed. Note too that two separate infrastructure installations can be mutually complementary. Hence, for instance, without power data networks don't work. And if data networks don't work, the ability to communicate notice of failure and restore power can be compromised.⁸⁵

Buffers to vulnerability

Networks – whether for communications installations, wind farms or other – need space, not just for physical installations but also for the establishment of buffer protection zones.⁸⁶ In New Zealand, for instance, buffer zones protect the country's undersea electricity, communications and energy cables.⁸⁷ Likewise in Europe, for various kinds of infrastructure.⁸⁸ This means that the actual footprint of infrastructure is larger than that of a physical installation itself. In addition, national security considerations and schemes to protect nature may limit how close to the coastline networks can be built, meaning further competition for space. The combination of such buffer zones often squeezes certain infrastructure assets into the remaining small areas. With no buffer zones, it could happen that several of those installations suffer damage or outage at the same time, whether due to intentional or unintentional action, or natural peril events.

Policymakers have recognised this and in many parts of the world, including the EU and the US, regulations on the use and protection of space for marine infrastructure are currently under review.⁸⁹ Infrastructure accumulation risk can also build in onshore locations. Typically, however, these are of lesser scale as applicable laws are country specific. In maritime locations, infrastructure networks can cross international sea borders. To adhere to the regulations of different jurisdictions can make projects more complex to manage and operate. That a large share of critical infrastructure is privately owned heightens the challenge of managing the accumulation risk, as different parties need to coordinate their respective activities.

In addition, critical energy and communication networks are of high strategic interest economically and militarily, and are thus potential targets for hostile attack. The closer together the network installations are, the more attractive they are as a target for a hostile agent, as a pre-meditated action could compromise several assets in one hit.

When providing coverage for infrastructure assets, insurers should keep in mind that property and business interruption losses can originate from different sources, be those natural catastrophes, human interference or cyberattacks. Especially exposed are chokepoints where sub-sea cables and other infrastructure assets connect to land.

Related articles in SONAR

- “Overlooked exposures: emerging risk drivers for Life & Health insurers”, SONAR 2023
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- “Beirut explosion – dangers of hazardous materials”, SONAR 2021
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Risky bets – democratising financial information through social media

Social media has become entrenched into daily life. This brings many benefits, but also potential financial market and insurance sector risks.

Potential impacts

- Information overload at speed and scale can lead to irrational investment decisions, losses on assets and financial market volatility. These may have repercussions on insurers' profitability.
- Increased claims for D&O and credit and surety, in case of bank runs and fallout from a financial market crisis.
- A likely outcome of the above is more demand for cyber insurance from corporate clients.
- Social media can influence reputational and commercial liability risks, as well as social inflation.
- On the upside, insurers can use social media data to more accurately identify potential risks and tailor insurance products accordingly, and also to identify claim inconsistencies and fraud.
- Social media tools can also improve customer engagement by facilitating direct feedback.

Impact

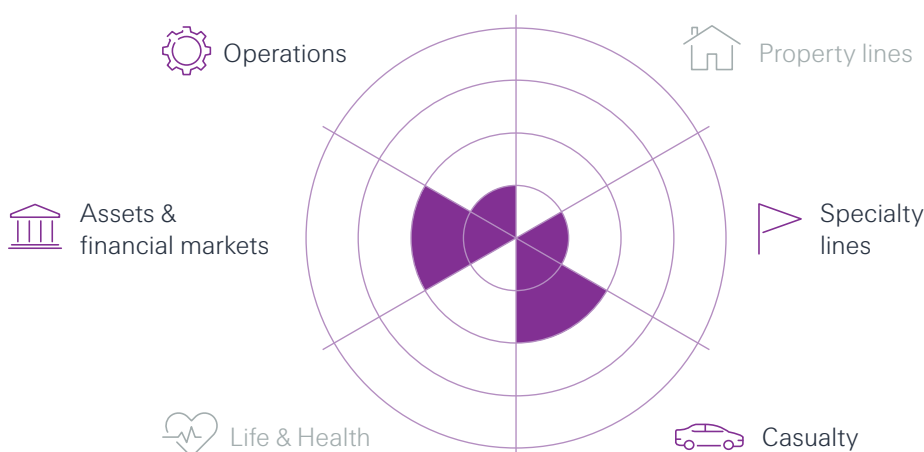


Time frame

0 – 3 years

> 3 years

Impacted business areas



Social media and financial markets

Social media (SoMe) platforms like X (formerly Twitter), LinkedIn and Reddit have democratised the flow of financial information. Individuals, regardless of their expertise, can now access financial market information in real time, and exchange ideas much more easily and inexpensively. As such, they are empowered to make more informed investment decisions.

The combination of social media (SoMe) and online stock trading has boosted the participation of retail investors by providing quick access to information and speeding up the process of purchasing and selling shares. In addition, there is the benefit of education: a wide range of resources available on SoMe platforms help lower the barriers to entry for retail investors. A recent report said that 34% of retail investors have

made at least one change to their investments in response to information garnered from SoMe.⁹⁰ Another study said that almost 249 institutional investors (out of 256) utilise SoMe for professional purposes, with 80% using it daily.⁹¹ Around 30% said that the information they gather on SoMe has had a direct impact on their investment recommendations and decisions.

SoMe platforms have also been used to spread misinformation (eg, "pump and dump" schemes⁹²) and to manipulate human behaviour. The risk of (mis)information overflow can impede effective decision-making and increase the likelihood of cognitive biases influencing investment decisions. False rumours and malicious intent can lead to significant (short-term) market volatility and cause financial losses for unsuspecting investors.

Herding at speed

The speed at which SoMe disseminates news does have disadvantages too. The collapse of the Silicon Valley Bank (SVB) in March 2023 was dubbed the world's first "Twitter-fuelled bank run".⁹³ Customers, prompted by viral panic on SoMe platforms, withdrew USD 42 billion in a single day from SVB, leaving it with a USD 1 billion negative cash balance. Bank runs like this typically lead to increased claims in D&O and credit and surety insurance.

SoMe platforms can also create "herd mentality", where investors follow popular trends without conducting proper due diligence. In 2021, the "wallstreetbets" community on Reddit caused prices of meme stocks (eg, GameStop⁹⁴) and cryptocurrencies to surge. While some members profited on the way up, many suffered heavy losses when the party ended (very abruptly). This has led to increased discussion on how securities regulation should be adapted to the modern era of investor behaviour and digital information exchange.⁹⁵

Social media benefits for the insurance sector

Insurers can use SoMe data to complement their traditional risk assessment methods, and to help identify new emerging risks. SoMe may also support tailoring and marketing of insurance products, and the identification of

claims inconsistencies and fraud (eg, in personal injury claims).⁹⁶ Estimates put the cyber insurance protection gap at around 90%.⁹⁷ Corporate demand for cyber insurance will likely rise given the increased potential for cyberattacks via SoMe channels.⁹⁸

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Technological and natural environment

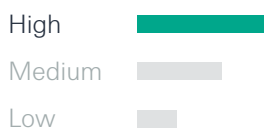
Beyond broken infrastructure – the cascading effects of natural catastrophes

Floods, wildfires, severe convective storms and other natural peril events routinely inflict widespread property damage and what can be massive, headline-grabbing economic and insurance losses.⁹⁹ Less well publicised and understood are the cascading (negative) effects of such events on the systems that underpin society, including energy, water and transport infrastructure.

Potential impacts

- The compounding effects of hazard occurrence and the interconnectedness of critical infrastructure and supply chains can generate loss accumulation.
- Where there is coverage, damage to critical infrastructure like transmission lines and power plants from natural peril events result in claims in property and business interruption insurance.
- Cascading effects of a natural disaster can include critical service outages, potentially generating claims for business interruption (for the costs of business closures, power outages, water contamination and transport issues), for contingent business interruption (from organisations reliant on uninterrupted power supply, eg, hospitals, data centres, security systems), property damage (eg, food spoilage due to non-functioning refrigeration), and for L&H and liability (eg, from impacts of power outages affecting medical facilities or water contamination). Insurers may also face operational disruptions.
- Accumulating and enduring impacts from natural catastrophes can hamper economic development, increase costs for the public sector, and slow insurance market growth.

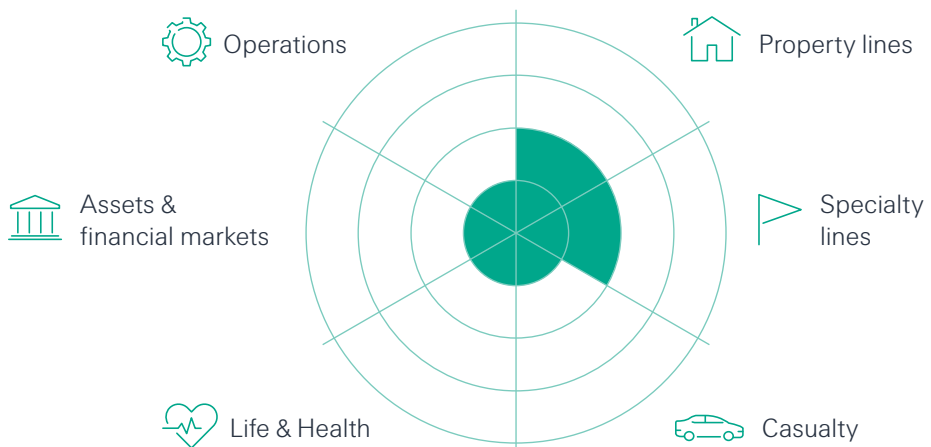
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



Water: from contamination to crisis

Wildfires leave a toxic legacy, polluting water and cutting access to it. Ash and debris contaminate water sources, requiring costly treatment and, potentially, water shortages. The wildfires in California in 2018 caused tens of billions worth of damages for infrastructure repairs, including of water systems and treatment facilities. Millions of Californians were affected by water-quality concerns.¹⁰⁰ Floods can compound negative outcomes, clogging rivers

and basins with sediment from eroded hillsides,¹⁰¹ inundating treatment plants and leaving communities vulnerable to disease outbreak.¹⁰² In the wildfire in Maui in 2023, damage to power lines knocked out pumps, leaving nearly 40 000 residents without water for days.¹⁰³

Wildfires and floods also threaten water quality by mobilising pollutants from hazardous waste sites.¹⁰⁴ The potential for widespread contamination poses a serious risk

to human health and the environment. Such concerns were prominent after the fire at the Iron Mountain Superfund site in northern California in 2018. The intense heat and altered water flows that followed exacerbated the threat of mobilising hazardous materials and disrupting waste water treatment, risking long-term contamination of water supplies and fisheries beyond the fire zone.¹⁰⁵

Energy grid: from blackout to breakdown

Imagine a wildfire scorching transmission lines, plunging a city into darkness. Not just an inconvenience; a lifeline severed. Hospitals scramble for backup generators, water treatment plants sputter, and communication networks falter. Similarly, substations submerged by floods can cripple entire regions, impacting healthcare, essential services and communications.

The economic impact? In 2021, winter storms plunged Texas into frigid temperatures and widespread power outages. These crippled critical infrastructure and disrupted business operations across various sectors, resulting in economic losses estimated to be more than USD 200 billion.¹⁰⁶ Impacts of disrupted generation and distribution include blackouts, damage to equipment and lost revenue. Production lines can be brought to a standstill due to lack of power, leading to lost production time, materials spoilage and delays to deliveries. Retail businesses may also close or operate at reduced capacity, leading to lost sales and revenue. Additionally, refrigerated goods can spoil without proper temperature control.

Transport: from road blocks to economic stagnation

Floods cripple the interconnectivity of transportation networks, causing local disruptions that ripple into regional travel issues and hinder emergency response systems. This can have significant economic consequences, as seen in

the 2021 Germany floods, where road and railway infrastructure damage cost an estimated USD 2 billion, with long-term closures further impacting mobility and hindering economic recovery.¹⁰⁷ Wildfires melt asphalt and damage bridges, severing vital economic arteries. The 2021 Dixie Fire in California impacted over 8 000 miles of roads, disrupting businesses and livelihoods. Hail can create hazardous driving conditions too, impacting millions of commuters and jeopardising essential deliveries.

Related articles in SONAR

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Cyber-enabled fraud – a new era for organised crime

Fraudsters and criminal groups can buy “crime-as-a-service” products online, thereby scaling their criminal activities with new technology. The costs, monetary and others, can be huge. All are at risk.

Potential impacts

- Operational fraud losses may be directly monetary (eg, if criminals hack business emails that involve money-transfer instructions), or through business-line claims fraud (eg, if criminals create fake emails from a hospital to healthcare insurers, requesting reimbursements for the provision of fictitious care treatment/services).
- Depending on the circumstance of a fraud incident, insurance covers for liability, cyber etc. may be triggered.
- Large fraud losses but also investment in preventive measures can increase operational costs, which insurers may in turn pass on to consumers by raising premiums.
- A firm could suffer reputational damage if found not to have invested enough in cyber infrastructure security.
- A large loss of funds can erode trust in the providers of financial services and lead to market volatility.

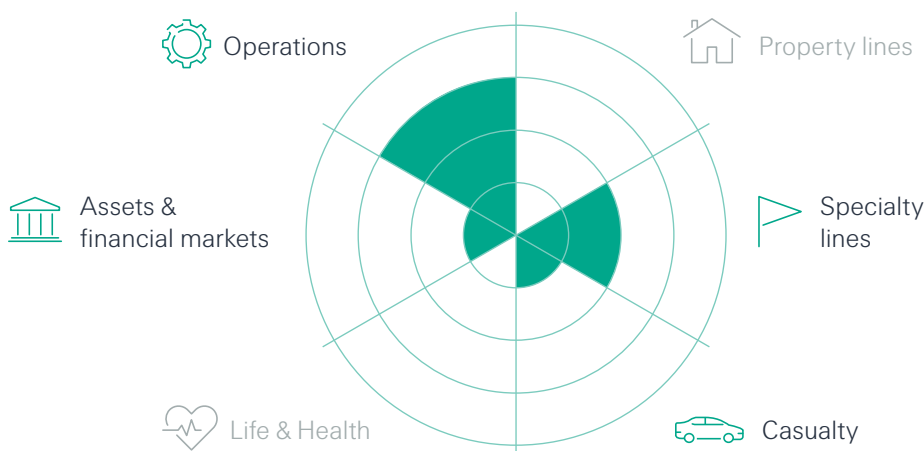
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



Organised crime goes digital

Organised criminal groups profit from several interconnected activities like trafficking of drugs, humans, arms and other illicit goods.¹⁰⁸ Technology allows criminals to connect, and also to reach consumers. Another main profit earner is cyber-enabled fraud, which no longer requires high technical acumen: crime-as-a-service products or “business models” can be bought online.¹⁰⁹ Cyber-enabled fraud does not depend on computers per se, but is enabled and scalable with new technologies. This is different from cyber-dependent crime like hacking and malware.¹¹⁰

Business email chains are a prime target for cyber fraudsters. They follow the communication in real time and wait for the right time to penetrate. This can be done

in different ways, for instance by contacting a paying party just after a claim has been wired. The message is to say the payment did not come through, and the ask for the paying party to resend the same amount to a different account number.¹¹¹ More advanced fraudsters can act as middle men, controlling the email traffic both ways and editing payment instructions.¹¹² So-called “threat actors” can also add personal information derived from earlier messages when tracking an email chain, adding a sense of (false) trust.¹¹³

Last year, Commonwealth Bank of Australia said it registers about 85 million possible cyber fraud events a day, while JP Morgan says it is seeing an increase in the number of cyber hacker attempts daily, and that it employs more

engineers than Google or Amazon to fight fraudsters.¹¹⁴ IBM stated in a report the average cost of a data breach is USD 4.45 million and that of a mega-breach USD 332 million.¹¹⁵ Beyond hacking emails, other types of cyber-enabled fraud are phishing, social media and mobile impersonation fraud, and employment scams. Illegal call centres have made cyber fraud scalable.¹¹⁶ In the summer of 2022, INTERPOL issued a warning that cyber fraud is escalating rapidly across the world, with online scam centres initially concentrated in Cambodia but since also appearing in other countries in southeast Asia, South America, East Africa and Western Europe.¹¹⁷

Think, check and act vigilantly

In the US, the FBI estimates that the total cost of domestic non-health insurance fraud could exceed USD 40 billion per year, equivalent to an average increase in annual premiums for one family of between USD 400 and USD 700.¹¹⁸ Like other providers of financial services, insurers manage large money transactions when helping clients recover losses from accidents, catastrophes and other adverse events.

As if all the above is not enough, a new era is already upon us: the use of Artificial Intelligence (AI) in cyber fraud. Deepfakes – artificial images and videos created by highly sophisticated AI tools – are now being used by organised groups.¹¹⁹ An example is bringing together an audio clip from social media or corporate video with an AI voice-synthesizing tool, for the purpose of staging a fake video call online, supposedly from a company's CFO.

With many avenues for cyber fraud, in addition to robust cyber security infrastructure, companies should provide regular (and mandatory) awareness trainings for employees and conduct regular fraud risk assessments. Whenever a payment instruction or data-sharing request in an email or

other online communication tool from a third-party is received, whether at home or at work, the response should be to think, check and act vigilantly. As part of corporate risk management, there should be a clear process for when it is OK to action payments, and easy access to internal fraud or compliance teams when in doubt.¹²⁰

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Alluring abyss – deep-sea mining’s murky future

Extracting minerals from the seabed is a new industry in the making, one that may well soon be operational. However, the unknowns and potential risks are many.

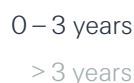
Potential impacts

- Deep-sea mining comes with many unknowns in terms of the technology used, operational risks, the effect on the environment and ecosystems, economic viability and regulations.
- There could be negative impacts on natural environment/ecosystem services and associated industries. Another outcome could be greenhouse gas emissions, which could trigger liability claims.
- For insurers, risk analysis throughout the deep-sea mining value chain is a challenge.
- Operators of deep-sea mining projects will likely need to have insurance cover as a regulatory pre-condition in order to be granted licenses. However, as of today, there is no market in deep-sea mining insurance.
- The complexity of deep-sea mining and associated supply chains may result in accumulation of business interruption exposures.
- Local communities may not support official approvals of projects, nor the granting of licenses for operation.¹²¹ Protests could add to business interruption risk.
- Current lack of a comprehensive and internationally consistent regulatory framework creates significant risks. Property & Specialty insurers may be impacted by uncertainty around regulation, with cases of unexpected litigation leading to also unexpected losses.
- Those who see deep-sea mining as an investment proposition have adopted a wait-and-see approach.

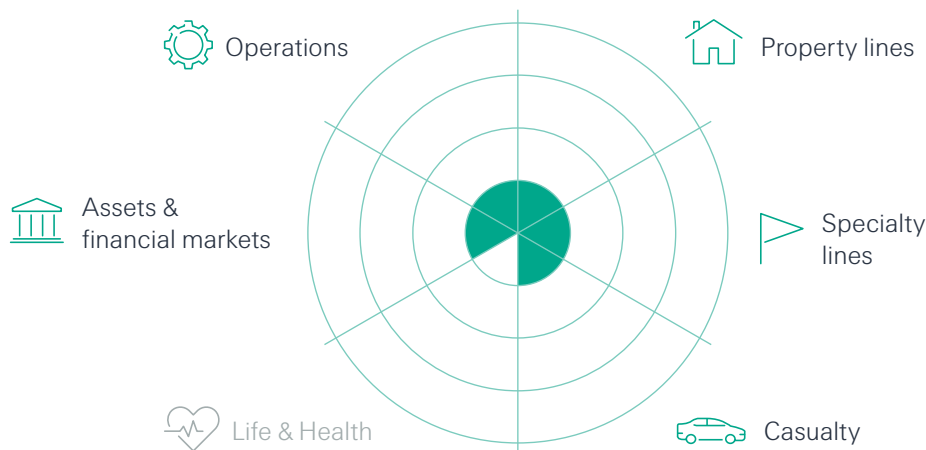
Impact



Time frame



Impacted business areas



Hunger for critical mineral resources

Some hail deep-sea mining – the extraction of metals and minerals from the ocean floor (200 metres/656 feet below the surface) – as the solution to society’s insatiable quest for materials such as cobalt, graphite and lithium.¹²² These are essential components of batteries, solar panels and other low-carbon technologies. A recent report says the supply of key minerals for emissions-reduction technologies may run short by 2030.¹²³

Hope that deep sea mining (DSM) will fill the gap and contribute to carbon reduction is thus understandable. It should not be overlooked, however, that DSM can also negatively affect carbon sequestration and large deep-ocean carbon sinks. When sea beds are disturbed, they can release significant amounts of carbon stored in the sediment into the water.¹²⁴ This undermines the transition benefits of DSM and may be a source of large liability risks.

Gaps in research and regulations

Some nations have applied for exploration permits from the UN's International Seabed Authority (ISA). After failing to reach an agreement in 2023, the ISA seeks to finalise DSM regulations by 2025, but many issues remain unresolved.¹²⁵ Either with explicit permits granted or without explicit prohibition, it could be that DSM projects are initiated soon.¹²⁶ But voices from science and environmental protection sound the alarm about current state of affairs. The call from scientists, NGOs, private companies¹²⁷ and also some nations¹²⁸ is for a moratorium on the roll-out of DSM projects until confirmed scientific information is available to enable informed decision-making and the establishment of robust sustainability standards.¹²⁹

Abyssal plains – a vast universe of biodiversity

Deep sea grounds provide a habitat for a largely yet unknown but unique biodiversity that form the basis of the deep-sea food chains.¹³⁰ Disruption to/destruction of biodiversity ecosystems caused by mining operations could have long-lasting or even irreversible impacts,¹³¹ with negative cascading effects on nature and related industries (fishing, ocean tourism etc.).¹³²

In 2022, all nations agreed that 30% of the world's oceans should be legally protected by 2030. As of today, only 3% has been so-designated.¹³³ If DSM projects are approved for operation in the not-too-distant future, it could be that some ecologically important areas are destroyed before legal protection is in place.

Technical and economic challenges abound

The main focus area of DSM is on harvesting polymetallic nodules (clumps of minerals and metals) through remotely operated vehicles (ROVs). Hydraulic pumps, bucket systems, large mobile infrastructures and supply chains are required to enable material flow. The complexity of operations and related emerging risks could come with many vulnerabilities, and potentially high costs.¹³⁴ It is not yet clear that DSM will be scalable and commercially viable.¹³⁵ Competition and the maturation of threat scenarios will depend on future regulation and international agreements. Viability will also be shaped by technological innovation and the development of alternatives to energy transition beyond the current critical minerals.

Related articles in SONAR

- “No room left – competition for space generates infrastructure accumulation risks”, this edition (SONAR 2024)
- “Recycling – when energy infrastructure becomes hazardous”, this edition (SONAR 2024)
- “Burning questions – risky lithium-ion batteries”, SONAR 2020
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Recycling – when energy infrastructure becomes hazardous

Solar panels, wind turbines, batteries and other infrastructure are the face of the green energy transition. At end of life, however, recycling these complex structures and materials is not so easy.

Potential impacts

- The risks relating to recycling of infrastructure assets will impact casualty lines of business most.
- The disposal process can lead to claims under product, general and environmental liability insurance when incidents with hazardous substances lead to property damage, bodily injury and financial losses.
- Recycling operations can involve hazardous substances and processes. In case of explosions, fires, or exposure to hazardous substances, employees can be harmed. This could spark worker compensation and employer liability claims.
- Disposal in landfills can lead to groundwater and soil contamination, giving rise to environmental liability claims.
- Exports of materials/structures to countries with lower and/or not enforced safety standards for disposal may violate the future EU Directive on Corporate Sustainability and Due Diligence. This could be source of D&O claims.
- The hazardous nature of the materials can also lead to property losses.

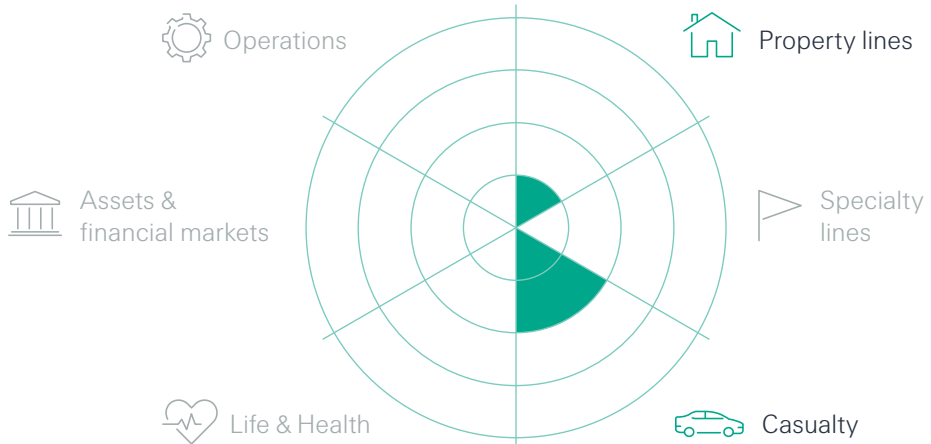
Impact



Time frame

0 – 3 years
> 3 years

Impacted business areas



Around 1.7 million tonnes of solar panels are expected to be discarded by 2030, and up to 60 million tonnes by 2050 globally.¹³⁶ In the EU alone, 25 000 tonnes of wind generator rotor blades will reach end-of-life by 2025. Neither the solar panels nor the rotor blades that will come to end-of-life in the next two decades were designed with recycling or disposal in mind. The potential for accidents involving hazardous materials or others will become an issue for insurers in the years ahead.¹³⁷

The responsibility for the safe disposal of these infrastructures will lie across value chains from the original

producers, to the operators and to the agents in charge of repair and disposal. Therefore it may be difficult to apportion the liabilities resulting from incidents that cause property damage, bodily injury or financial loss. This ambiguity will become important for insurance claims.

The EU and other countries/regions are now moving in the direction of a circular economy, with a requirement that products be designed for sustainable disposal.¹³⁸ However, solar panels and rotor blades have previously been designed for performance. This has meant use of composite materials technologies, whereby components

are glued, baked or via other means closely connected for best performance.¹³⁹ The problem is that it is very difficult to take components apart at end-of-life.¹⁴⁰

Technologies to recycle renewable energy infrastructure are still under development. This creates a disconnect between the push for renewable energy for climate reasons and compliance with the demands of a circular economy. How regulators will resolve this is not clear.

Disassembling the physical frames for solar panels and towers for windmills for recycling is relatively straightforward. But the photovoltaic cells within the solar panels and the rotor blades are amalgams of many materials. There are currently two main processes to take these apart: pyrometallurgy¹⁴¹ and hydrometallurgy.¹⁴² Pyrometallurgy requires the component to be heated to high temperatures with little or no oxygen present. Hydrometallurgy involves leaching the components with solvents, such as strong acids or other hazardous compounds. Both processes carry risks. High temperatures can trigger fires and potentially explosions, leading to property and casualty losses. Hazardous substances can injure and kill people, and damage physical assets. The result could include property, casualty and environmental liability losses.

Inadequate disposal of panels or blades can contaminate the soil, water and air, which in turn can trigger claims for bodily injury, property, environmental and/or financial losses. And where waste is exported and inadequately disposed of, this could violate future regulations like the EU Corporate Sustainability Directive.¹⁴³ If such a violation becomes public, the share price of listed companies involved could come under pressure. This may spark D&O claims.

When insurers look to offer coverage to corporates, they should investigate the firms' risk management practices. Technologies have developed fast in recent years¹⁴⁴ and the waste and other products that need to be recycled may present different levels of hazard risk. Insurers should consider that current recycling capacity and capabilities do not match demand, and that new players are entering the market. New facilities designed with safety procedures in mind are needed. This is true for windmills, solar and other technologies like batteries and fuel cells used for the green energy transition.¹⁴⁵ Recycling will be important for the energy transition and a circular economy, and close cooperation between insurers and recyclers can provide means forward for its insurability.¹⁴⁶

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¹⁴⁵ M. Massoud et al., [Review on recycling energy resources and sustainability](#), Heliyon 9, April 2023.

¹⁴⁶ See White Paper by Swiss Re Institute, in cooperation with AXA, BCG IKA, INGKA, [What goes around comes around: Insuring the circular economy](#)



Next generation nuclear

Innovation in nuclear technology, in particular the development of Small Modular Reactors (SMRs) has potential to secure safe energy for all. Still, the build-up of new nuclear technologies at scale, especially SMRs, faces challenges that demand international cooperation and policy vision.

Recent innovation in nuclear technology promises significant benefits in reducing operational hazard potential, progressing net-zero ambitions, contributing to energy independence and security of supply, and stimulating economic growth. In this context, SMRs, a new generation of reactors with a power capacity of a few MW(th) to 300 MW(e), represent a significant departure from the larger nuclear power plants (NPPs) that have been in operation to date. They offer diverse options for advanced nuclear power with a wide range of capabilities, power output and applications, including heat and power generation. They can be land- or marine-based, stationary or mobile.

Nevertheless, the promise of the new faces challenges, such as the complexity and cost of developing needed supply chains, coping with shortages of appropriately skilled workers and navigating the heterogeneous international regulatory environment. Furthermore, nuclear waste disposal will likely always remain a challenge.

Deglobalisation trends offer the nuclear industry an opportunity to relocate long-lead items manufacturing, and promote domestic value creation. In the near-term though, current inflation pressures, and disruptions in the labour, capital, raw material and nuclear fuel supply markets, may complicate the deployment of new technologies, particularly SMRs, as seen in the US recently.¹⁴⁷ In addition, increased exports of nuclear goods, reduced reactor sizes and increased mobility would require commensurate

measures to establish advanced safeguards and strong control mechanisms to address potential proliferation challenges and terrorist threats.

Moreover, legal considerations like the extent of coverage under the nuclear liability conventions for diverse types of SMRs will likely give rise to new questions.¹⁴⁸ Current conventions will be challenged by innovative logistics solutions needed to secure the deployment of SMRs globally. On the insurance side, the new technology developments will call for cradle-to-grave solutions that have yet to be invented and that respond to SMR's specific needs, covering all of the production, transport, installation, operation and decommissioning phases of the value chain.

Today's nuclear landscape is one of geographical disparity. Russia and China are in the lead, already operating fourth-generation NPPs, SMRs and expressing strong willingness to export their know-how. On this front, the West lags well behind. New nuclear programmes have yet to scale, to the extent that the first SMRs currently being developed by actors ranging from start-ups to incumbent nuclear energy providers, are unlikely to be in operation before the end of this decade.

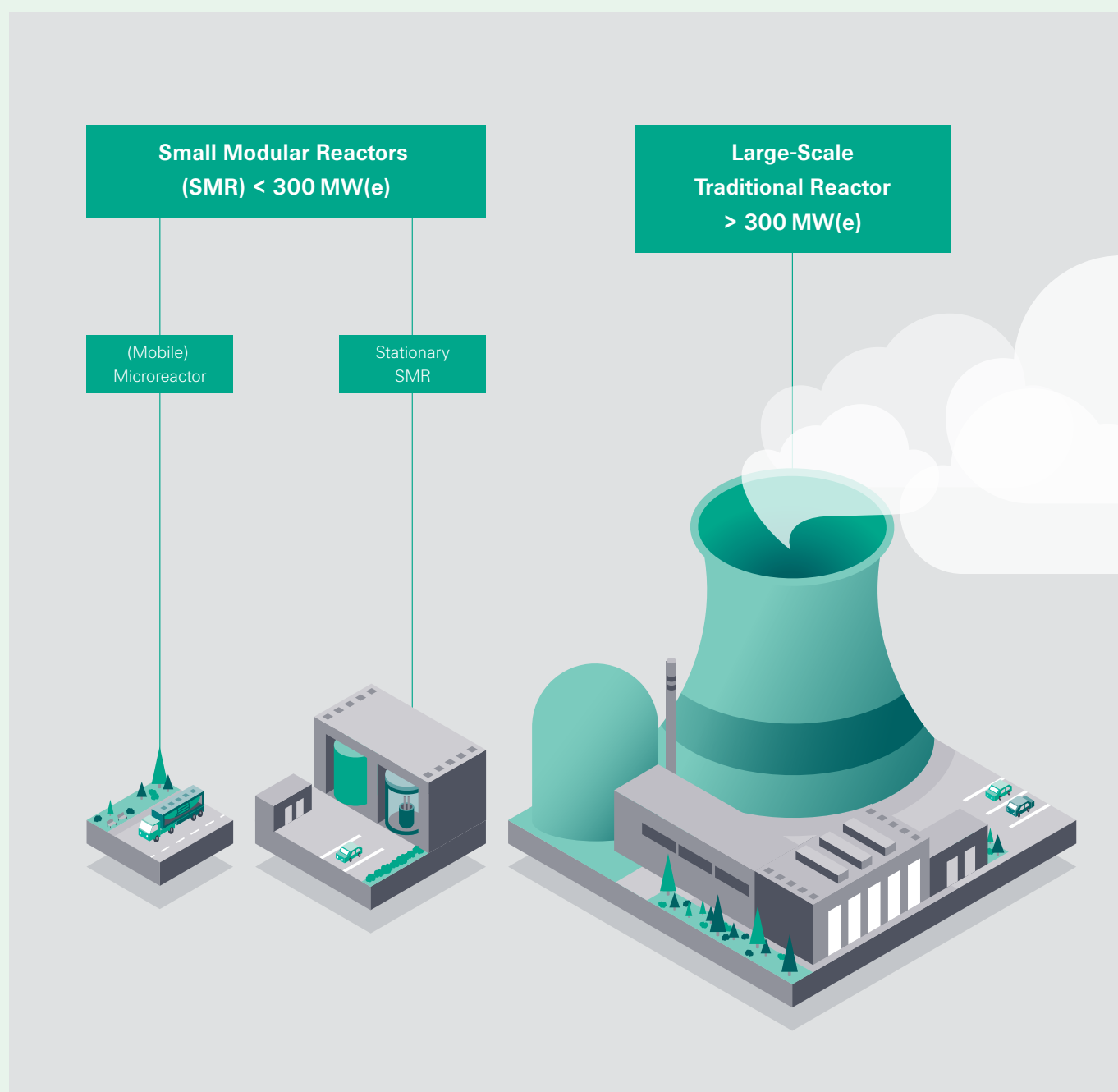
Catalysts for successful development and deployment of SMRs and other innovations will include policymaker acknowledgment of the strategic importance of nuclear energy. It will also require initiatives to encourage private

capital investment. Such as, for instance, the recent inclusion by the European Union of nuclear energy production in the list of economic activities covered by the EU taxonomy for sustainable activities.¹⁴⁹ Insurers' knowledge will be especially beneficial for start-ups new to nuclear, enabling better risk understanding, informed decision-making and increasing the bankability of their offering.

¹⁴⁷ D. Dalton, [NuScale Cancels Idaho Small Modular Reactor Project as Costs Increase](#), NucNet, 9 Nov. 2023.

¹⁴⁸ V. Roland, [Applicability of the existing nuclear liability conventions to different types of small modular reactors currently under development](#), Nuclear Law Bulletin, No. 110, 2023/1.

¹⁴⁹ European Commission, Directorate-General for Financial Stability, Financial Services and Capital Markets Union, [EU taxonomy: Complementary Climate Delegated Act to accelerate decarbonisation](#), 2022. This inclusion has not come without controversy, and environmental campaigners are currently suing the European Commission over its decision to include gas and nuclear in an EU guide to "green" investments, arguing it violates EU climate law and undermines efforts to achieve carbon neutrality by 2050. The inclusion of gas and nuclear is contested due to concerns about their environmental impact, with a judgment expected in 2025.







Competitive and business environment

AI – unintended insurance impacts and lessons from “silent cyber”

The increasing use of artificial intelligence (AI) could trigger claims across many lines of business. Insurers will need to develop an understanding of intended and unintended effects, and design products that mitigate the risks.

Potential impacts

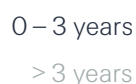
Different AI-related events could impact various lines of insurance business.

- Operational shutdowns resulting from AI-system malfunctions could trigger business interruption claims.
- Professionals could face claims for (1) erroneous advice or misinterpretations; and (2) unexplainable decisions delivered by AI-driven research or other tools that have negatively impacted end users.
- Manufacturers of AI-enhanced products could be subject to property damage and/or bodily injury claims resulting from AI failures or malfunction, or where product liability regulations have been violated.
- Corporate leaders could face accusations of failing to oversee/mitigate the risks associated with implementation of AI-driven processes that have led to financial losses or reputational damage. D&O and liability claims may be triggered.
- AI-driven hiring practices that inadvertently introduce bias could trigger discrimination lawsuits and claims against employers for unfair employment practices.
- Copyright violations and/or patent infringement as a result of leveraged training data or from the AI model itself could result in claims in liability coverages.
- Increased use of AI in healthcare diagnostics could change insurance demand, but also give rise to potential gaps in coverage.
- Insurers could face claims increases due to erroneous advice or misinterpretations delivered by AI-driven underwriting tools. AI-driven underwriting models that inadvertently introduce bias could trigger discrimination lawsuits and claims against insurance companies.

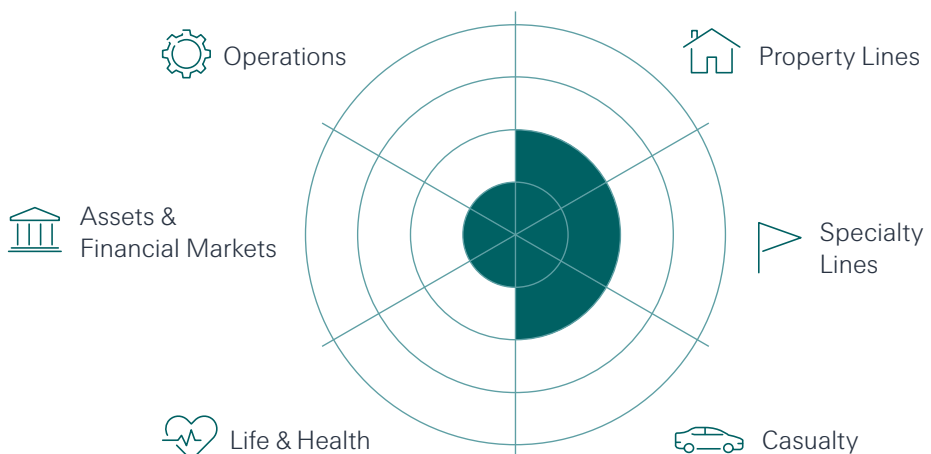
Impact



Time frame



Impacted business areas



AI to revolutionise business

AI has changed the way companies operate and will revolutionise global business practices further.¹⁵⁰ For example, generative AI that produces text, images, videos or other outputs, comes with a multitude of benefits, but also risks.¹⁵¹ And, as the number of AI incidents rises rapidly,¹⁵² so too is C-suite level awareness of the dangers.¹⁵³

AI is set to transform the insurance industry across various dimensions.¹⁵⁴ Enhanced underwriting, AI-supported customer services, claims processing automation and predictive analytics for fraud detection, are just a few examples of the potential benefits. All told, however, the main utility of insurance remains risk transfer, in this case as protection against a variety of risks that their clients may face through their own increasing use of AI.

AI risks: use traditional or new insurance policies?

Harm caused by AI can be “material or immaterial, including physical, psychological, societal or economic”.¹⁵⁵ It is unlikely that a single insurance policy will cover all potential risks that AI presents. As of today, AI risks are neither explicitly mentioned, limited nor excluded in policy language, and different exposures may be covered by different policies already in existence. Insurers need to analyse how AI-risks are dealt with in existing policies, and how to best deal with them in the future. Which risks are already covered, and which perhaps only silently so (in other words unintentionally, due to perhaps ambiguous language). It could be that the latter require alternative costing and risk assessment approaches to those inherent in existing policies, and/or even completely newly designed insurance products.

Lessons learned from “silent cyber” to be incorporated into insurers’ AI activities

With silent cyber, the industry has already learned lessons. There have been instances where some risks have been covered in non-cyber line policies, even though that was not the intention. With silent AI, it is time to prevent repetition of the same mistakes by understanding which risks traditional policies already (silently) cover. With the fast development of AI and associated regulations,¹⁵⁶ some of today’s assumptions may turn out to be wrong or incomplete. However, that should not prevent discussions around silent AI from starting now.

Understanding the impact of AI events on traditional policies

A scenario-based approach can help build understanding of AI risk use-cases. The scenarios and their impacts can be assessed following a structured process and set of questions, to understand how current policy wording/coverages would

apply to specific risk cases. With these insights, insurers will be better placed to devise AI-risk transfer solutions that match customers’ future protection needs.¹⁵⁷

Related articles in SONAR

- “AI getting hacked – systemic vulnerabilities of a booming technology”, SONAR 2023
- “Digital twins: virtual product testing and maintenance”, SONAR 2023
- “Generative AI”, SONAR 2023
- “Legal tech – is AI rendering justice?”, SONAR 2022
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- “Wiggle room – Artificial Intelligence and healthcare”, SONAR 2019
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¹⁵⁰ [Can AI help you solve problems?](#), McKinsey, 21 May 2023.

¹⁵¹ D. Hendrycks et al., [An Overview of Catastrophic AI Risks](#), arXiv:2306.12001 (cs), 9 Oct. 2023; J. Anderson and L. Rainie, [Experts Predict the Best and Worst Changes in Digital Life by 2035](#), Pew Research Center, 21 June 2023.

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¹⁵³ [In 2024, Generative AI Will Transition From Hype To Intent \(forbes.com\)](#), 6 Nov. 2023.

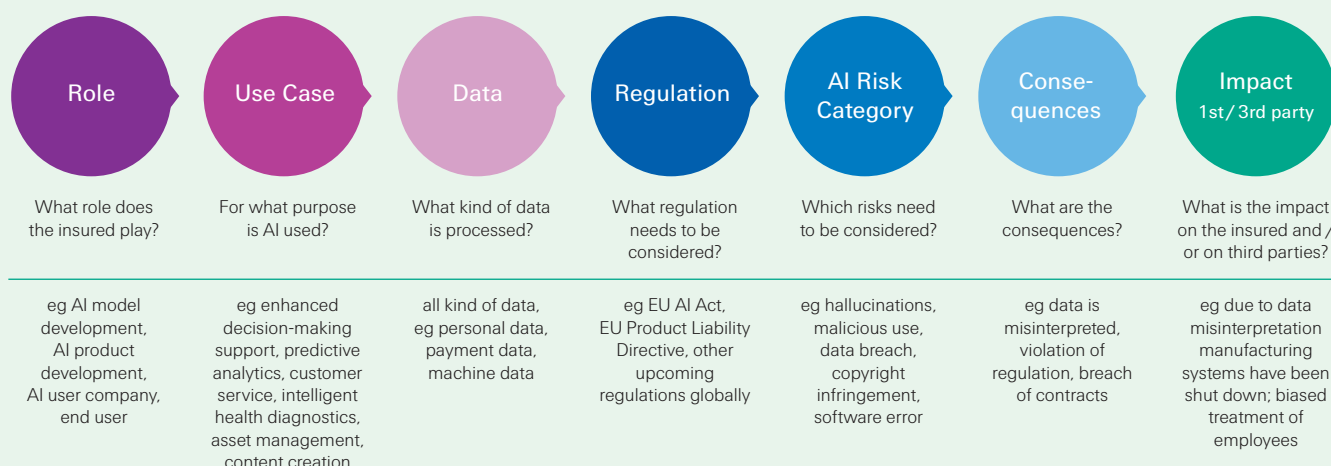
¹⁵⁴ P. Ladva and A. Grasso, [Benefits and use cases of AI in insurance, Part 2: Decrypting AI for insurance](#), Swiss Re, 17 April 2023; R. Balasubramanian et al., [Insurance 2030 – The impact of AI on the future of insurance](#) | McKinsey, McKinsey, 12 March 2021; L. Chordas and J. Weber, What the AI Revolution Means for Insurance Prospects, Policyholders and Regulators, AM Best’s Review, Nov. 2023; L. Chordas and J. Weber, How AI Is Remaking Insurance Organizations, AM Best’s Review, Nov. 2023.

¹⁵⁵ [EU Parliament and the Council of the EU, Proposal \[AI act\], 2021](#).

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¹⁵⁷ A new Swiss Re Institute study assesses AI risks and insurance potentials across industry sectors: Tech-tonic shifts: How AI could change industry risk landscapes, Swiss Re Institute, May 2024, [AI and the industry risk landscape | Swiss Re](#).

AI-Induced Scenario map: high-level scenario development



How do policy wordings respond?



Insurance fairness: challenge and opportunity

Developments such as climate change and digitalisation are changing risk pools and can challenge the viability of traditional insurance products. Risk-adequate pricing may affect insurance affordability and raise questions around fairness. Transparency, technology and insurance innovation can offer solutions.

Occurrence of more intense weather-related catastrophes (eg, tropical cyclones, floods, wildfires) could challenge the insurability of certain perils in specific regions,¹⁵⁸ potentially necessitating higher premium rates or even a termination of certain risk covers.¹⁵⁹ This can cause discontent among insured populations. Some homeowners may find it unfair that they are being charged higher premiums to cross-subsidise the losses on other homeowners' high-risk properties.¹⁶⁰ Others may perceive it as unjust that they may not be able to afford insurance from one year to the next.¹⁶¹ Such perceptions of inequity are likely boosted by a general trend of settlement in high-risk areas.¹⁶²

Risk balance issues and fairness concerns are being raised in other fields of insurance too. Within Life & Health (L&H) insurance, the digitalisation of our lives and breakthroughs in machine learning (eg, generative AI)¹⁶³ and novel diagnostic data (eg, from wearables or tests), potentially enable greater granularity and accuracy of risk quantification.

Self-monitoring and ongoing data collection can enable us to better track individual health, including abnormal readings which may indicate the onset of ill health. This in turn may help us live longer, healthier lives.¹⁶⁴ Early detection of cancer, for example, can improve chances of recovery and lower associated rates of mortality.¹⁶⁵ However, the proliferation of such technologies may also lead to information asymmetry and anti-selection risks. Consumers may have knowledge of their health status based on information acquired by over-the-counter diagnostics tools. Should they not, where adverse, disclose this information to insurers, it could lead to non-transparent subsidisation of their elevated risk status by other policy holders, and could even render certain risks uninsurable.

Amid these developments, the re/insurance industry potentially faces accusations of unfair practices. Such accusations may come from opposing sides in a given case, with diverging perspectives of fairness. In some instances, the underlying assumptions may be incompatible.¹⁶⁶ To address concerns around fairness is no small challenge. But debate and propositions around fairness and inequality have always been core to the world of insurance, and the concerns should be tackled. What can the insurance industry do to alleviate concerns, and are there recent examples?

Most productively, re/insurance companies transparently lay out their core capabilities of identifying risk pools and enabling innovative risk transfer solutions. Protection gaps need to be closed to reduce inequalities, and affordable insurance helps.¹⁶⁷ Core insurance values such as reliability, security and transparency remain vital for gaining the trust of clients, especially in the growing digital domain.¹⁶⁸

Among new approaches in Property & Casualty business to help make vulnerable populations more resilient in face of natural perils is parametric insurance. Recent examples include protection against severe weather conditions in high-growth markets.¹⁶⁹ Another example is a flood insurance scheme in New York that supports low- and mid-income households.¹⁷⁰ And yet another case is Flood Re, a public private partnership in the UK that incentivises adaptation and mitigation measures. Flood Re helps households and communities transition to risk adequate pricing gradually by keeping flood insurance affordable for a certain period of time.¹⁷¹ On the L&H side, alternative data from wearables could have the potential to enhance underwriting or enable dynamic underwriting. And digital platforms can improve access, affordability and service for customers, thereby making insurance more inclusive.¹⁷²

¹⁵⁸ "1 in 20 Australian homes could be uninsurable by the end of the century.", J. Moss and D. Burkett, [Social Justice and the Future of Fire Insurance in Australia](#), UNSW, Sydney, May 2020, p. 4.

¹⁵⁹ F. Khoo and J. Yong, [Too hot to insure – avoiding the insurability tipping point](#), FSI Insights on policy implementation, No 54, Nov. 2023.

¹⁶⁰ Flat Premiums can lead to regressivity of insurance i.e., the poor subsidizing the rich. S. Owen and I. Noy, [Regressivity in Public Natural Hazard Insurance: A Quantitative Analysis of the New Zealand Case](#), Economics of Disasters and Climate Change, 3, 2019.

¹⁶¹ California Department of Insurance, [Operational Assessment Report California FAIR Plan Association](#), June 2022.

¹⁶² J. Rentschler et al., [Global evidence of rapid urban growth in flood zones since 1985](#), Nature, 622, 2023.

¹⁶³ S. Germans et al., [Using sequences of life-events to predict human lives](#), Nature Computational Science 4, Ja. 2024.

¹⁶⁴ P. Patkee and A. Strange, [The future of life expectancy: Forecasting long-term mortality improvement trends for insurance](#), Swiss Re Institute, May 2023.

¹⁶⁵ P. Patkee et al., [Multi-Cancer Early Detection: Cancer screening beyond today's boundaries](#), Swiss Re Institute, Jan. 2024.

¹⁶⁶ P. Garg P. et al., [Fairness Metrics: A Comparative Analysis](#), IEEE, 2020.

¹⁶⁷ [Reshaping the social contract: the role of insurance in reducing income inequality](#), Swiss Re Institute, sigma, 3, 2022.

¹⁶⁸ [Decoding digital trust – An insurance perspective](#), Swiss Re Institute, May 2022. [Decoding digital trust II – A consumer perspective](#), Swiss Re Institute, May 2023.

¹⁶⁹ [Building resilience in India | Swiss Re](#).

¹⁷⁰ [A historic first for New York City](#), Swiss Re Public Sector Solutions, March 2023.

¹⁷¹ M. Cullen, Think Piece 8: Sharing risk or smoothing bad luck – what is insurance really all about?, in Association of British Insurers, [A brave new world: The changing landscape for insurance and long term savings](#), Nov. 2015, p. 40.; [Protecting Communities, Preserving Nature and Building Resiliency](#), California Department of Insurance, Climate Insurance Working Group, July 2021.

¹⁷² J. Schoonbee and A. Musnitzky, [Underwriting with alternative data 1: Principles to consider](#), L&H Trend Spotlight, Swiss Re Institute, Nov. 2022; J. Schoonbee and A. Musnitzky, [Underwriting with alternative data 2: Getting practical with wearables](#), L&H Trend Spotlight, Swiss Re Institute, April 2023.



Appendix: terms and definitions

What is SONAR?

SONAR stands for Systematic Observation of Notions Associated with Risk. It is Swiss Re's process for identifying, assessing and managing emerging risks. Experts across the company use a web-based platform to collect early signals of emerging risks. All signals are assessed and prioritised by an emerging risk management team, which works closely with topic experts from Swiss Re's different business areas. The team serves as a catalyst for risk identification and assessment to define and implement recommendations in collaboration with the business. The findings are regularly shared internally and summarised for external audiences here.

What are emerging risks?

We define emerging risks as newly developing or changing risks that are difficult to quantify and could have a major impact on society and industry.

What are emerging risk themes?

Emerging risk themes illustrate potential new or changing risk developments for the insurance industry. They are mainly derived from SONAR but also draw on other sources. All themes have been assessed and edited by Swiss Re's emerging risk management experts. This report only features new emerging risk themes (ie, topics covered in previous editions are not listed again). You can retrieve prior reports from our webpage: www.swissre.com/sonar.

What is meant by overall impact?

The overall impact of an emerging risk is an indicator of the potential financial, reputational and/or regulatory effect on the insurance industry. It is assessed on a scale from high to low:

- **HIGH**
Potentially high financial, reputational and/or regulatory impact or significant stakeholder concern
- **MEDIUM**
Potentially medium financial, reputational and/or regulatory impact or moderate stakeholder concern
- **LOW**
Potentially low financial, reputational and/or regulatory impact, or low stakeholder concern

What is meant by time frame?

We divide themes into those likely to occur in less than three years and those likely to occur over a longer horizon. This assessment should not be used as an indicator of when action is needed, as some themes likely to occur in the more distant future may, nonetheless, require immediate action to prepare.

What is meant by impact per business area?

Spider graphs indicate the potential impact on major insurance industry business areas on a scale from 0 (= no impact) to 4 (= significant impact).

What are trend spotlights?

Text boxes throughout the report provide selective focus articles on emerging trends which could become relevant for the insurance industry and its clients. The selection of topics is non-exhaustive, and descriptions are intended as food for thought and discussion starters rather than comprehensive reviews.

What are macro trends?

Swiss Re has identified a set of macro trends assumed to have a high impact on the insurance industry within the next five to 10 years. The macro trends featured in this report have been selected independently through expert discussions and surveys. They provide context to the emerging risk insights from the SONAR process.

Abbreviations

ADHD	Attention-deficit/hyperactivity disorder
AI	Artificial Intelligence
BI	Business Interruption
C&S	Credit and Surety
CFO	Chief Financial Officer
D&O	Directors and Officers
DSM	Deep-sea mining
GDP	Gross Domestic Product
ISA	International Seabed Authority (United Nations)
L&H	Life and Health
LSD	Lysergäurediethylamid
NHS	National Health Service (UK)
OOP	out-of-pocket (spending)
P&C	Property and Casualty
ROV	Remotely Operated Vehicle
SCS	Severe Convective Storms
SMR	Small Modular Reactor
SoMe	Social Media
UHC	Universal Health Coverage
UK	United Kingdom (Great Britain)
UN	United Nations
WHO	World Health Organization

Title

Swiss Re SONAR – New emerging risk insights
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